A taste of numerical hydrodynamics



Picture from Wilhelm Kley 2016

Role of simulations in astronomy

Astronomy is an observational science, we cannot perform physical experiments to gain understanding

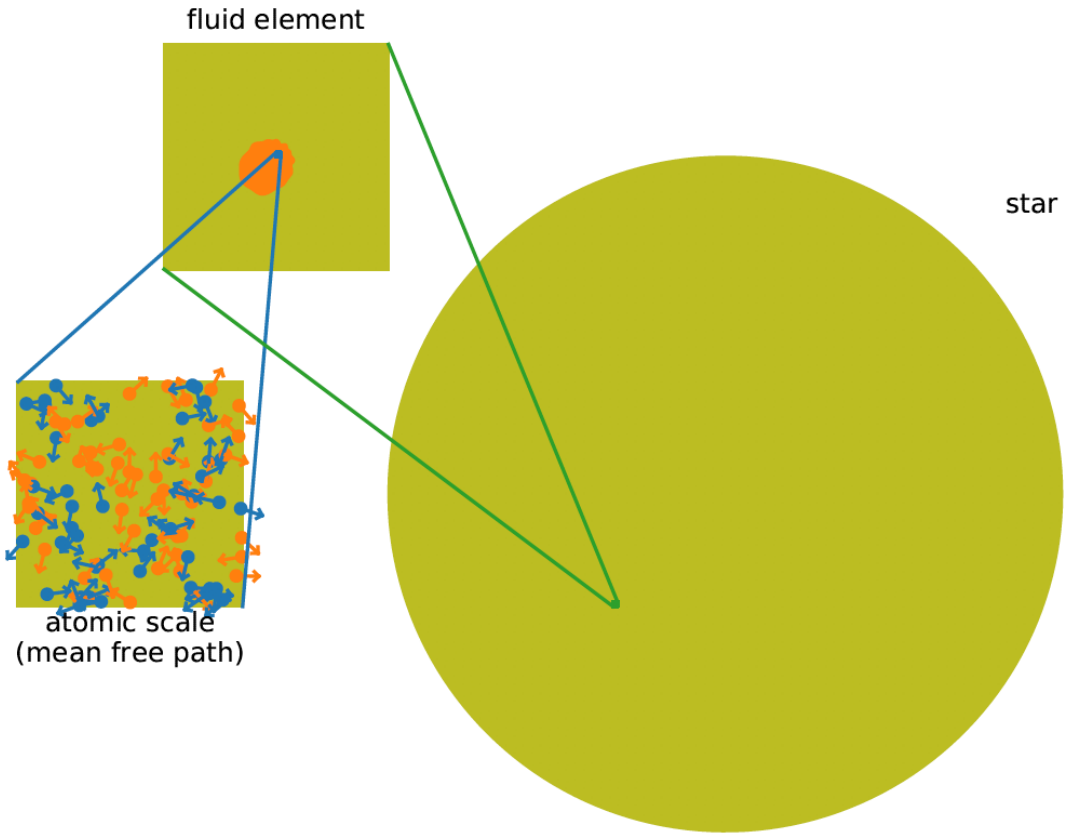
Numerical simulations, allows us to perform virtual experiments on systems

Understand how the system reacts to a range of different conditions

Predict the future evolution of physical systems

Fluid approximation

Macroscopic theory:
Fluid described as a continuum
Works on scales >> mean-free path



Basic Hydrodynamical Equations

Eulerian Form

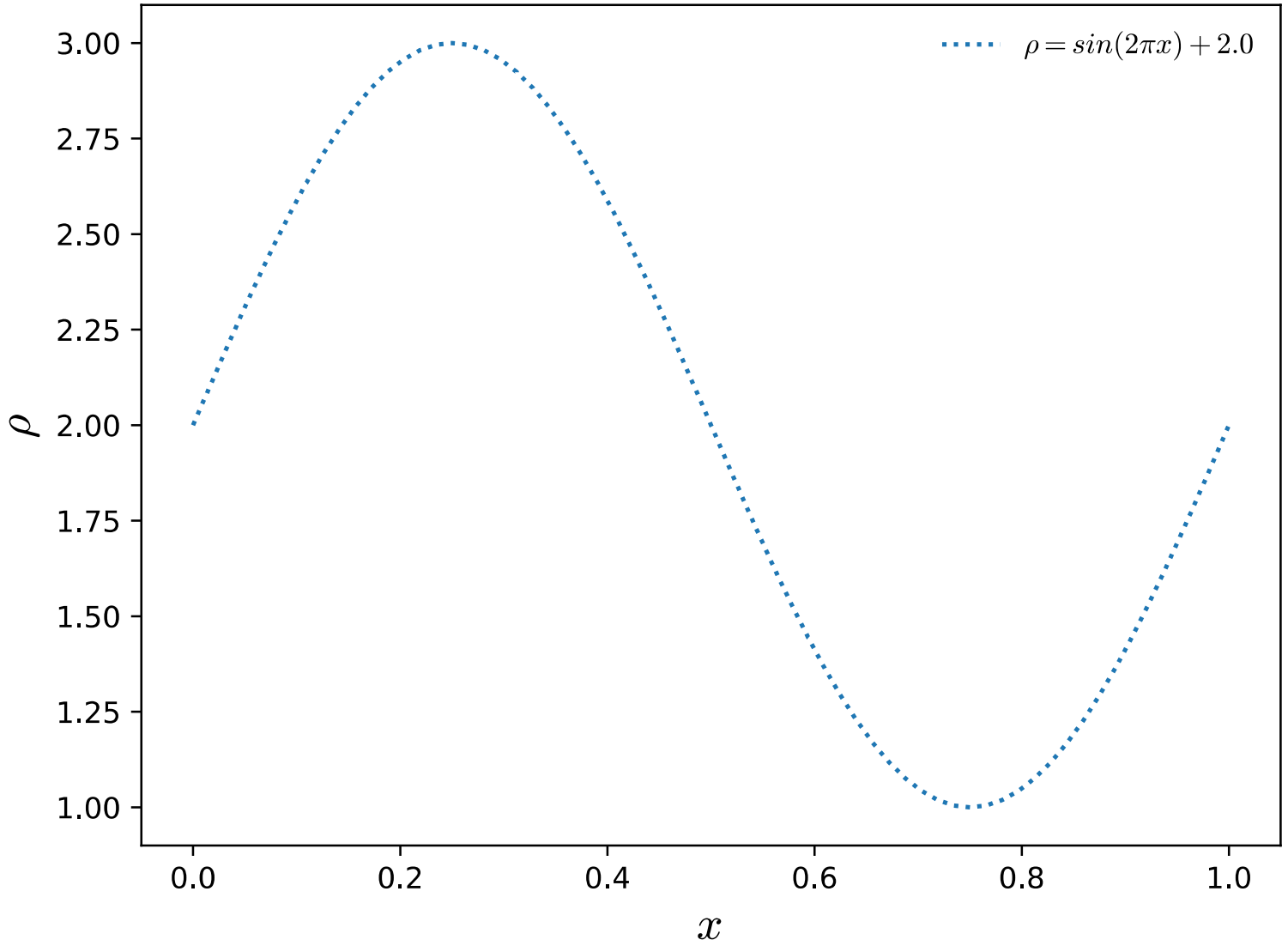
Continuity equation: $$\frac{\partial \rho}{\partial t} + v\nabla \cdot \rho = -\rho \nabla \cdot v$$

Equation of motion: $$\frac{dv}{dt} + v\nabla \cdot v = -\frac{\nabla P}{\rho}$$

Conservation of energy: $$\frac{du}{dt} + v\nabla \cdot u = -\frac{P}{\rho}\nabla \cdot v$$

Equation of state: $$P = (\gamma - 1)\rho u$$

A simple 1D density wave



1D system with

$$\rho = \sin 2\pi x + 2.0$$

$$v = v_x \hat{\boldsymbol{x}}$$

$$P = const$$

$$u = const$$

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Continuity equation:

$$\frac{\partial \rho}{\partial t} + v\nabla \cdot \rho = -\rho \nabla \cdot v$$

$\rho \nabla \cdot v$ = 0

Equation of motion:

$$\frac{dv}{dt} + v\nabla \cdot v = -\frac{\nabla P}{\rho}$$

$v\nabla \cdot v$ = 0, $\frac{\nabla P}{\rho}$ = 0

Conservation of energy:

$$\frac{du}{dt} + v\nabla \cdot u = -\frac{P}{\rho}\nabla \cdot v$$

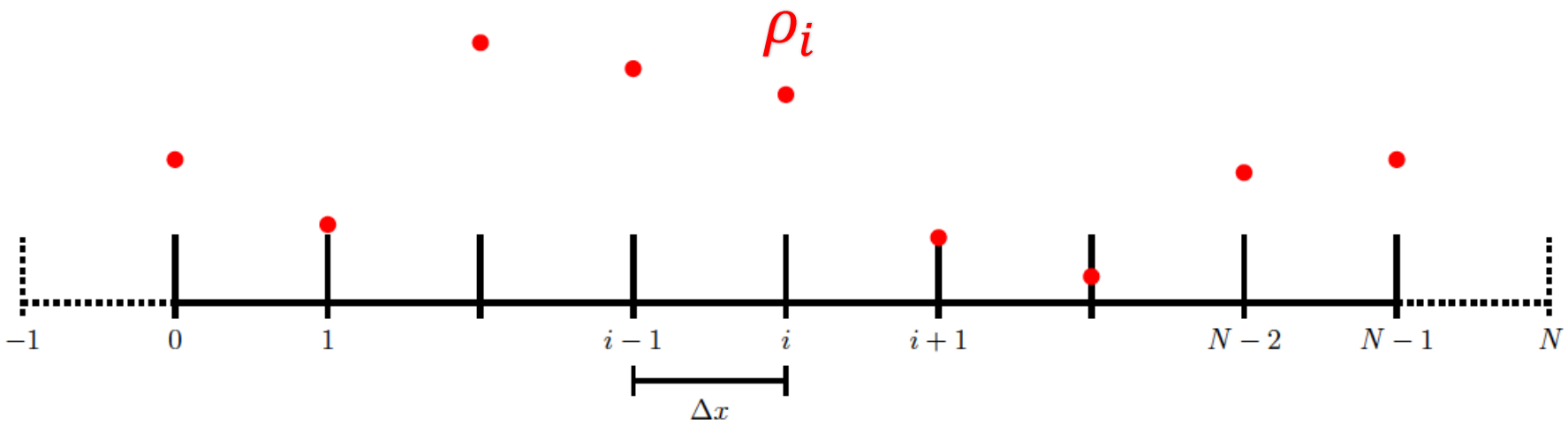
$v\nabla \cdot u$ = 0, $\frac{P}{\rho}\nabla \cdot v$ = 0

Advection equation in 1D ($v_x = const$)

$$\frac{\partial \rho}{\partial t} + \mathrm{v}_x \frac{d\rho}{dx} = 0$$

Discretization of the continous fluid in 1D.

Setup simple 1D grid, quantities(ρ, v, P, u)
stored at equally spaced points



Advection equation in 1D ($v_x = const$)

$$\frac{\partial \rho}{\partial t} + \mathrm{v}_x \frac{d\rho}{dx} = 0$$

Determining derivatives

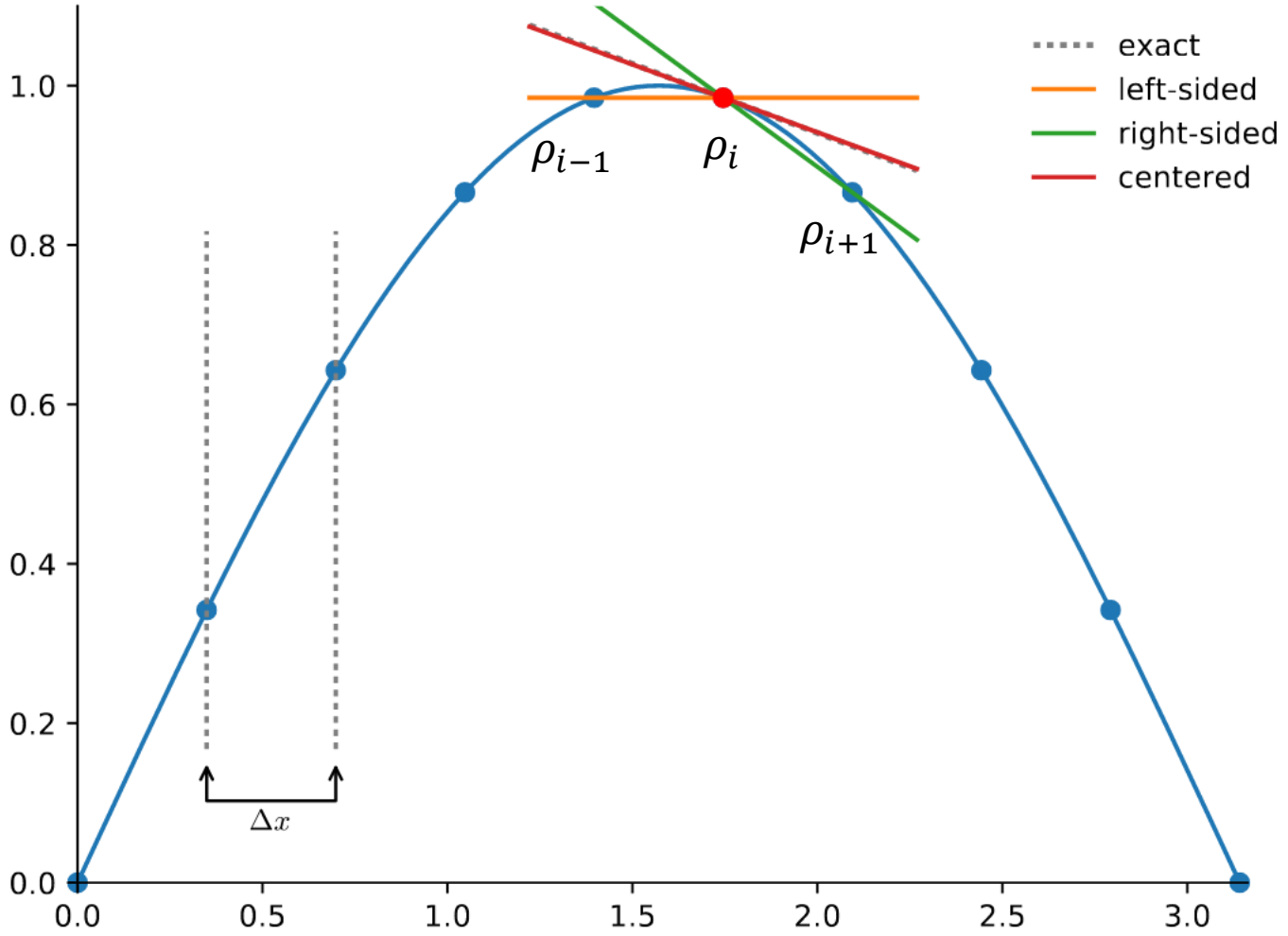
One sided difference

$$\frac{d\rho}{dx} = \frac{\rho_i - \rho_{i-1}}{\Delta x} + O(\Delta x)$$

Centered difference

$$\frac{d\rho}{dx} = \frac{\rho_{i+1} - \rho_{i-1}}{2\Delta x} + O(\Delta x^2)$$

$O(\Delta x)$ indicates the leading order error



Advection equation in 1D ($v_x = const$)

$$\frac{\partial \rho}{\partial t} + \mathrm{v}_x \frac{d\rho}{dx} = 0$$

Spatial derivative

$$\frac{d\rho}{dx} = \frac{\rho_i^n - \rho_{i-1}^n}{\Delta x}$$

Time derivative

$$\frac{\partial \rho}{\partial t} = \frac{\rho_i^{n+1} - \rho_i^n}{\Delta t}$$

Here n describes the current timestep

Inserting and reordering gives:

Advection equation in 1D ($v_x = const$)

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Inserting and reordering gives:

$$\rho_i^{n+1} = \rho_i^n - \frac{\mathrm{v}_x \Delta t}{\Delta x}(\rho_i^n - \rho_{i-1}^n)$$

First-order upwind scheme ($v_x > 0$)

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First-order upwind scheme ($v_x > 0$)

System only stable if timestep $\Delta t < \frac{\Delta x}{v_x}$

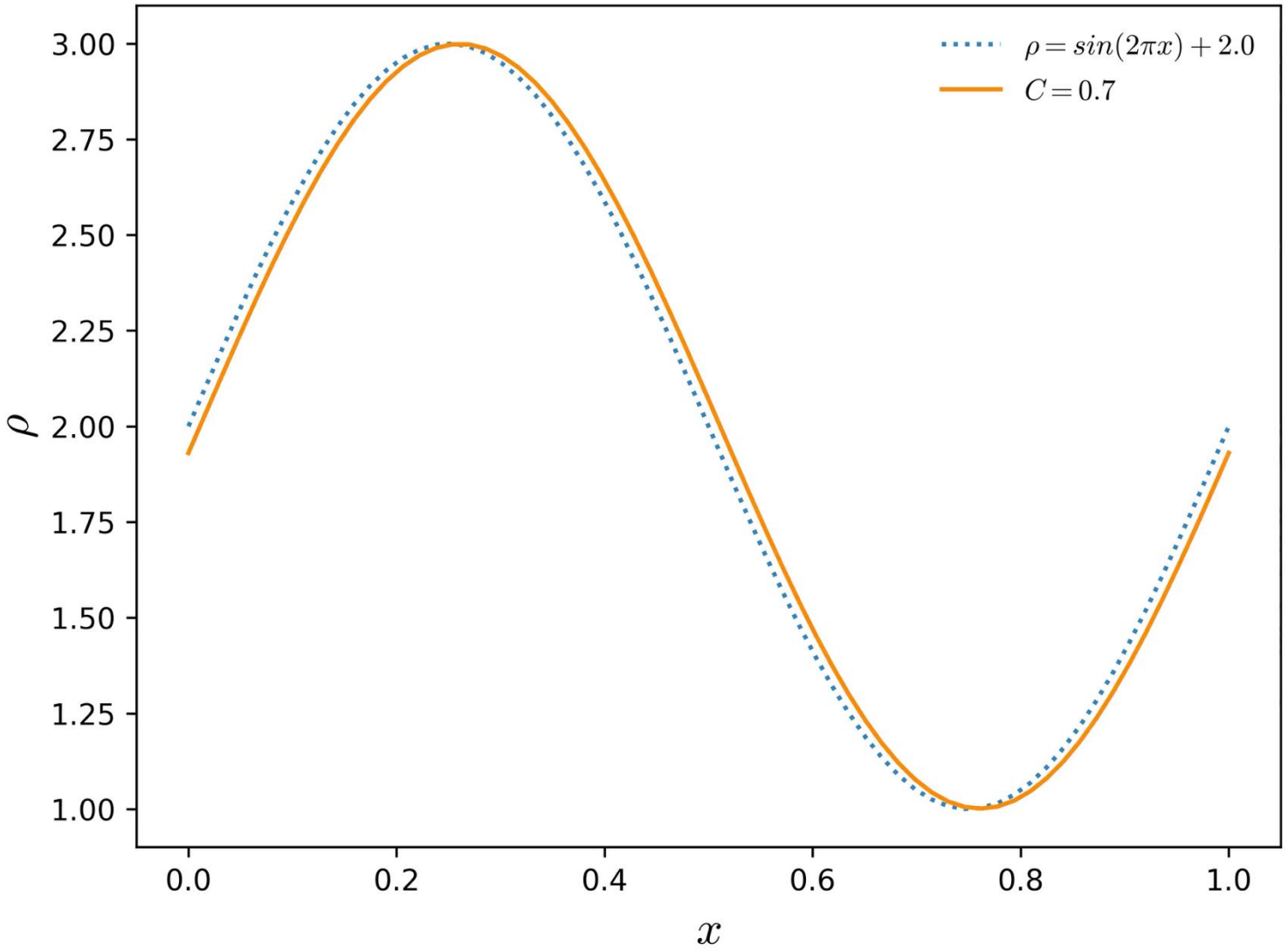
As for larger timesteps information propagates across more then a single zone

$$\Delta t = C \frac{\Delta x}{v_x} \qquad C < 1$$

C is the Courant-Friedrichs-Lewy(CFL) number

Typical value for standard simulations $C \approx 0.7$

A simple 1D density wave



1D system with

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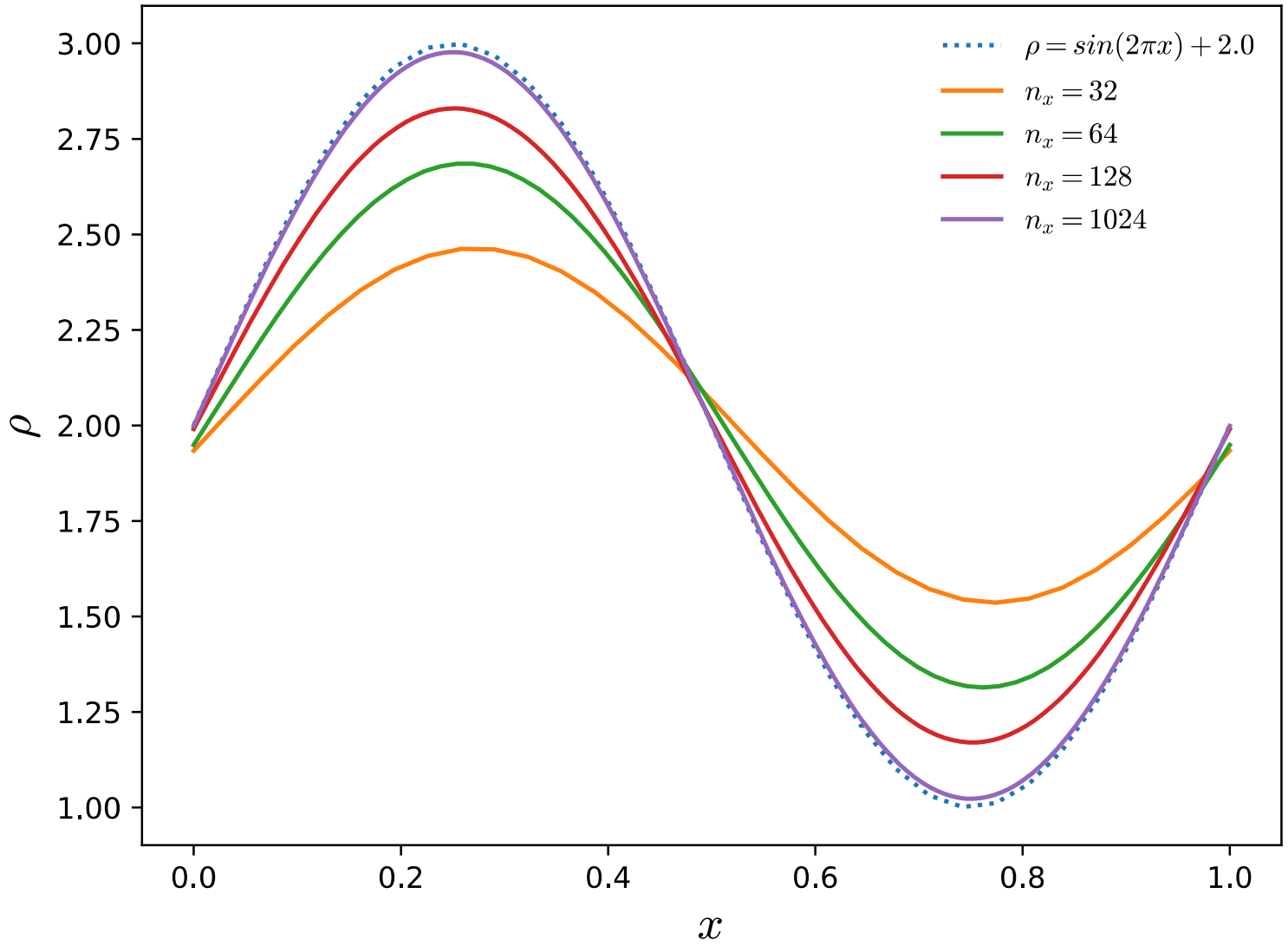
Resolution

$$n_x = 64$$

$$\Delta x = 1.0/n_x$$

$$\Delta t = 0.7 \frac{\Delta x}{v_x}$$

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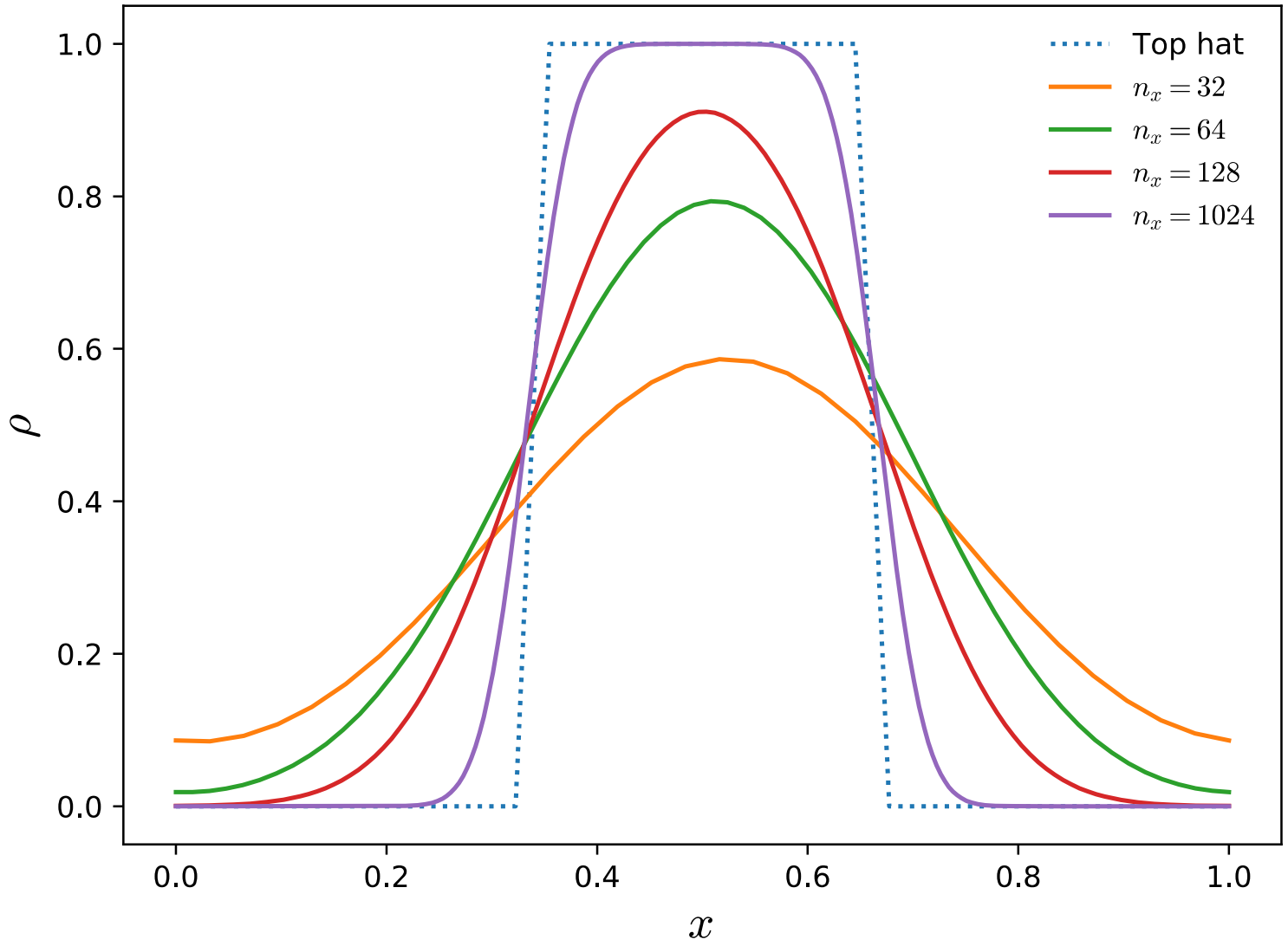
$$u = const$$

Resolution

$$\Delta x = 1.0/n_x$$

$$\Delta t = 0.7 \frac{\Delta x}{v_x}$$

A top-hat 1D density wave



1D system with

$$\rho = 1.0 \quad [1/3 < x < 2/3]$$

$$\rho = 0.0 \quad [outside]$$

$$v = v_x \, \hat{\boldsymbol{x}}$$

$$P = const$$

$$u = const$$

Resolution

$$\Delta x = 1.0/n_x$$

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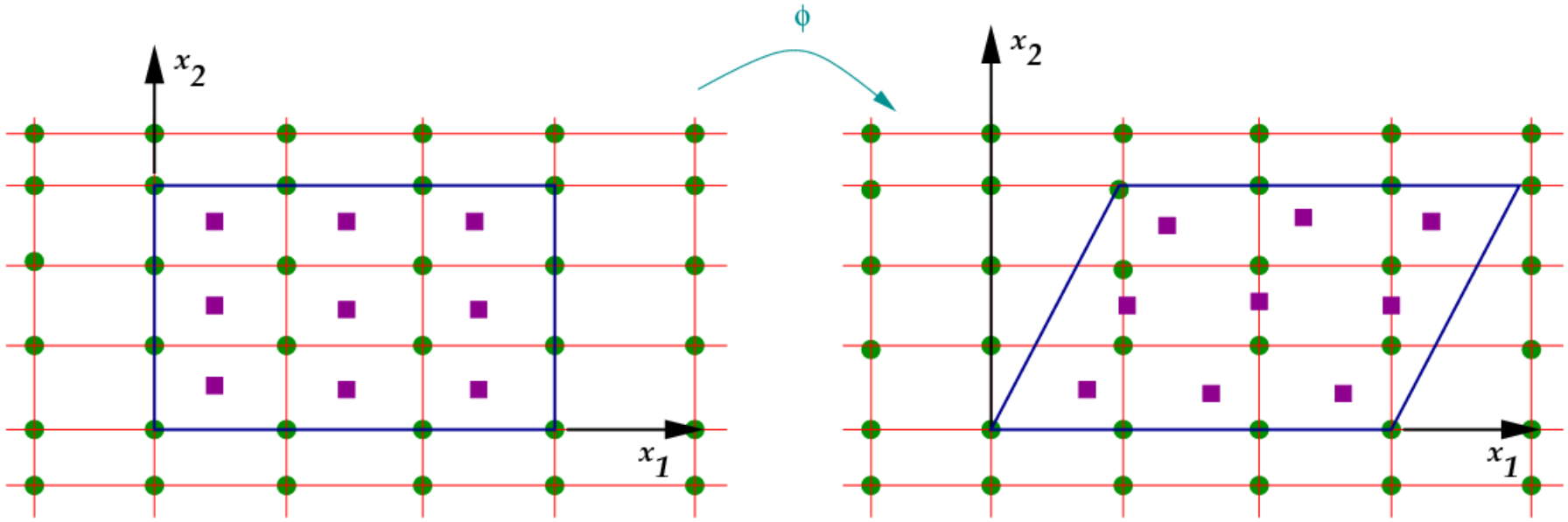
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Descriptions of Fluid Flow

$$\underbrace{\frac{d}{dt}}_{Lagrangian\ derivative} = \underbrace{\frac{\partial}{\partial t} + \boldsymbol{v} \cdot \nabla}_{Eulerian\ derivative}$$

- Eulerian

Coordinates fixed



Descriptions of Fluid Flow

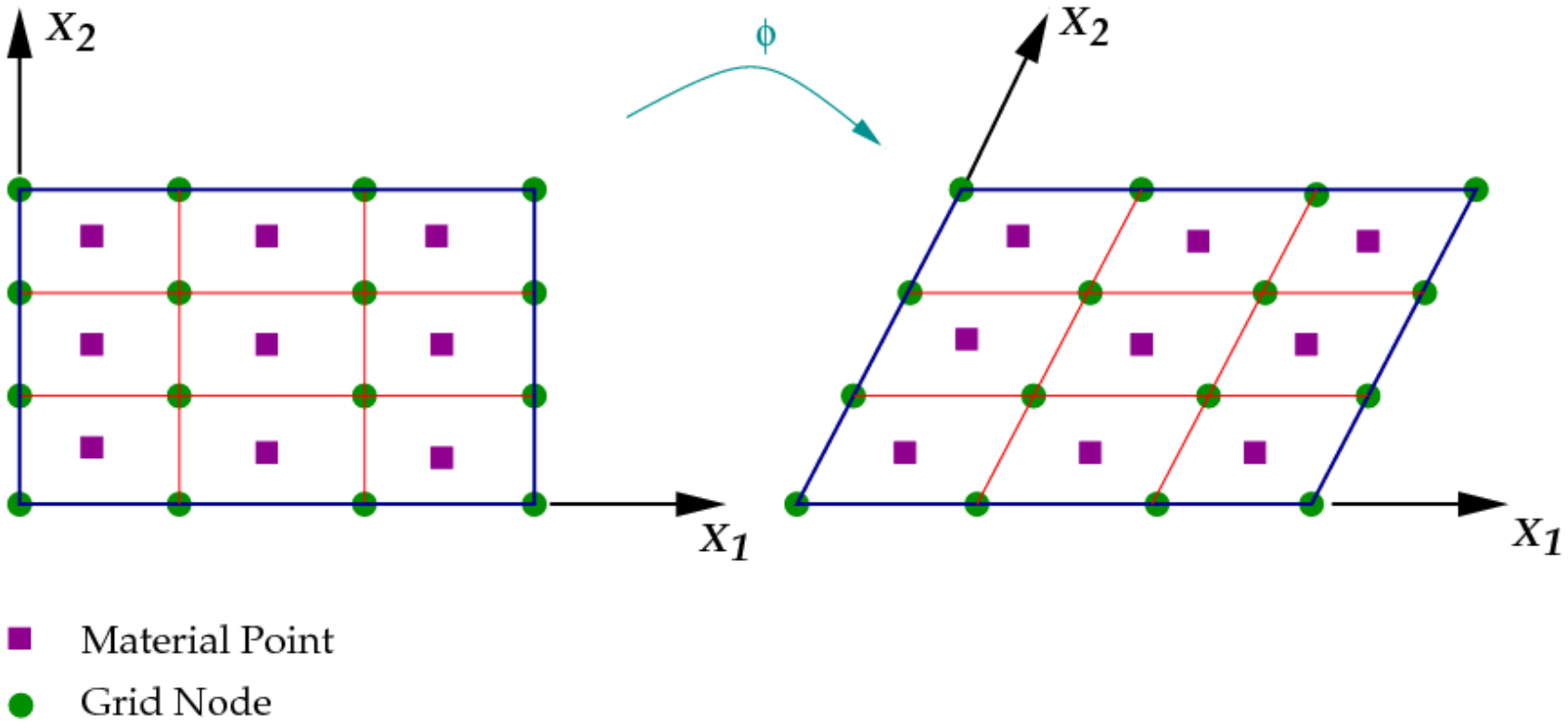
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- Eulerian

Coordinates fixed

- Lagranian

Coordinates move with flow



Basic Hydrodynamical Equations

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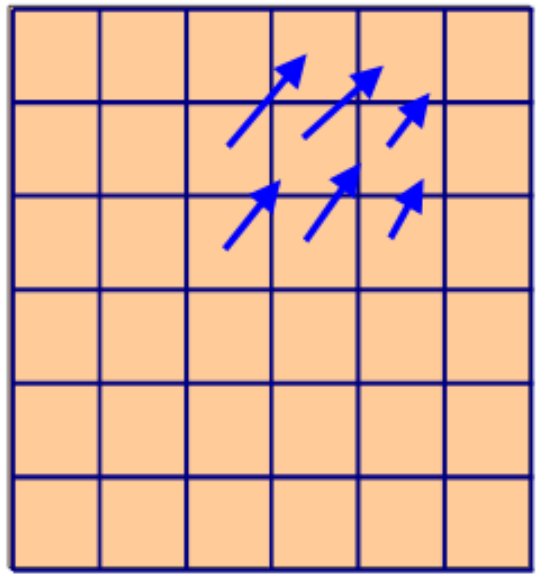
$P = const$

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Information within cells does not change

Coordinates/cells move with flow

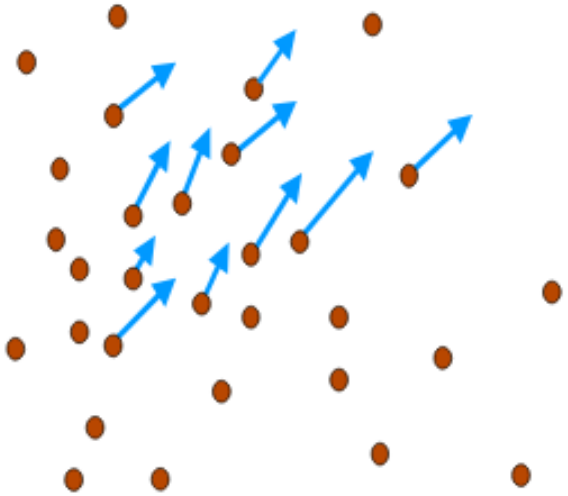
Discretizing the fluid



- Mesh based method

Discretize by volume

Fluid is represented on a mesh



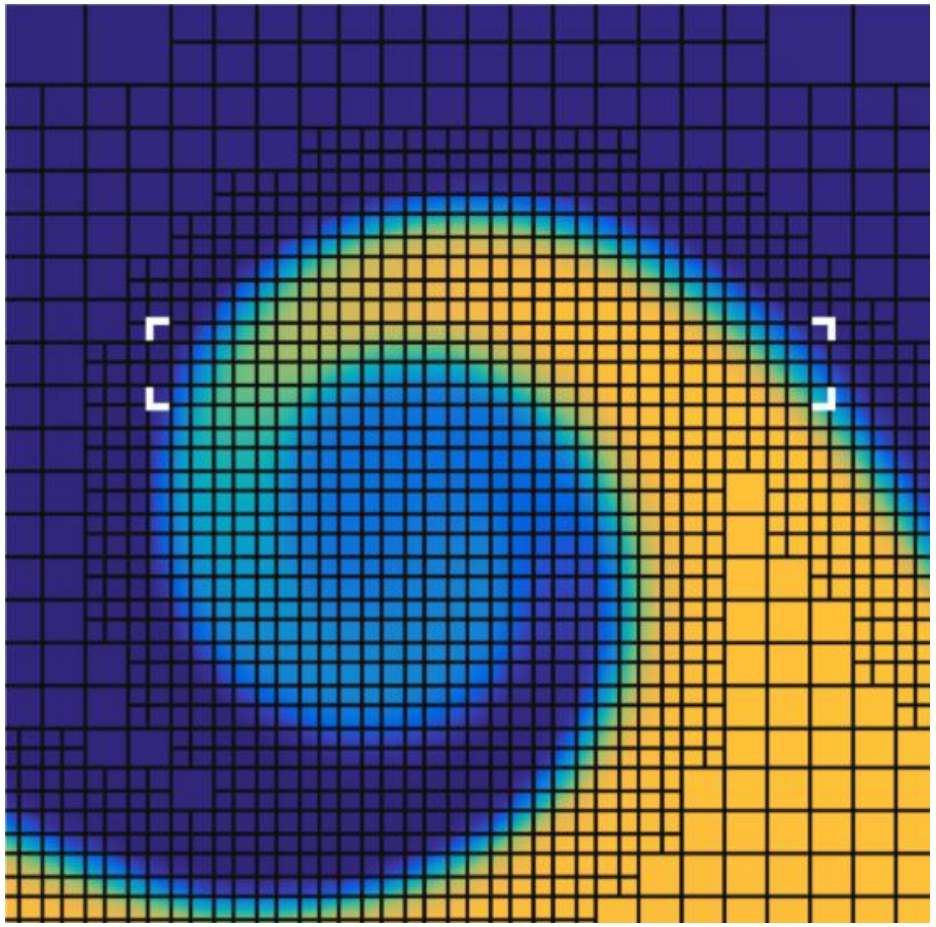
- Meshfree method

Discretize by mass

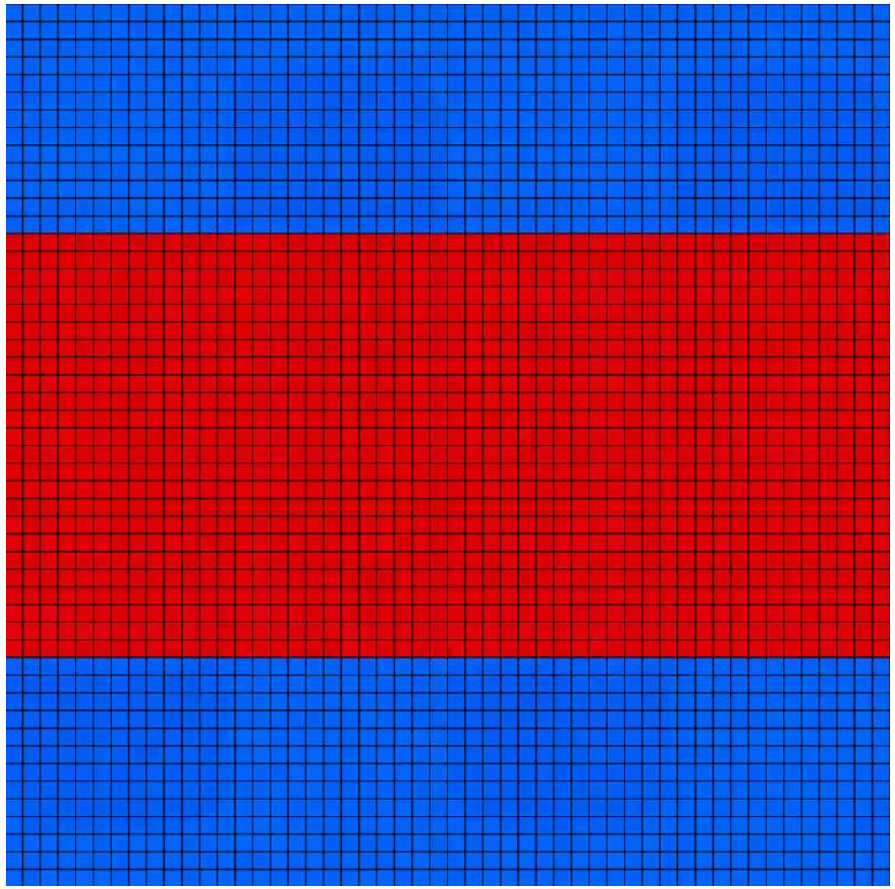
Fluid is represented with fluid elements

Mesh Method

- Eulerian



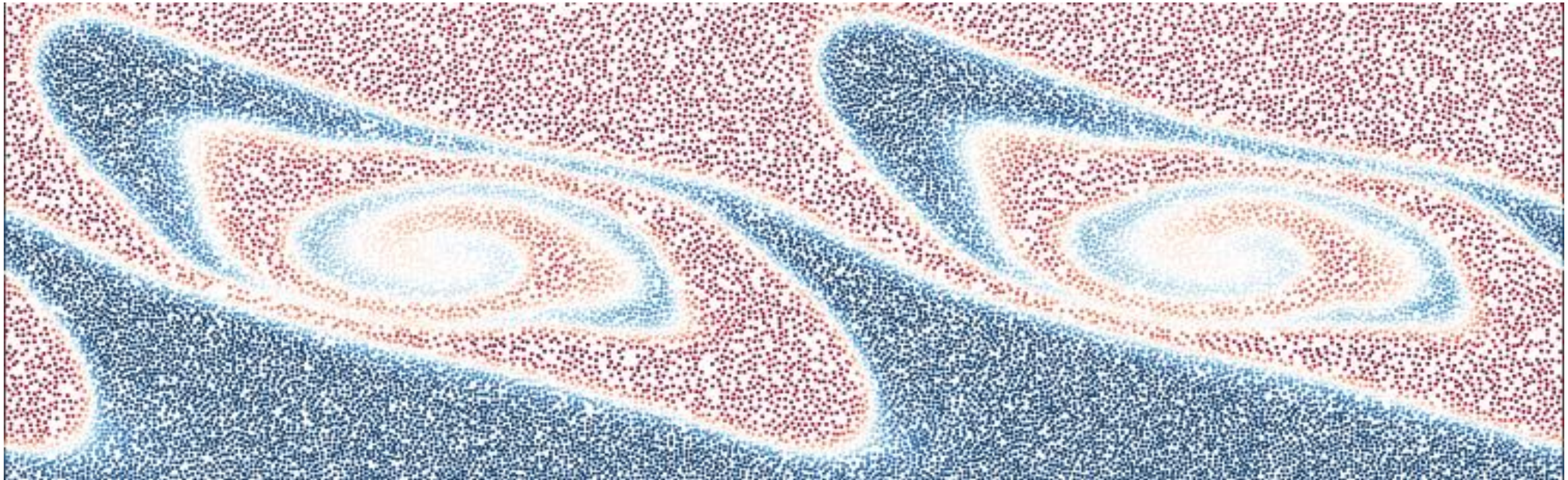
- Lagrangian

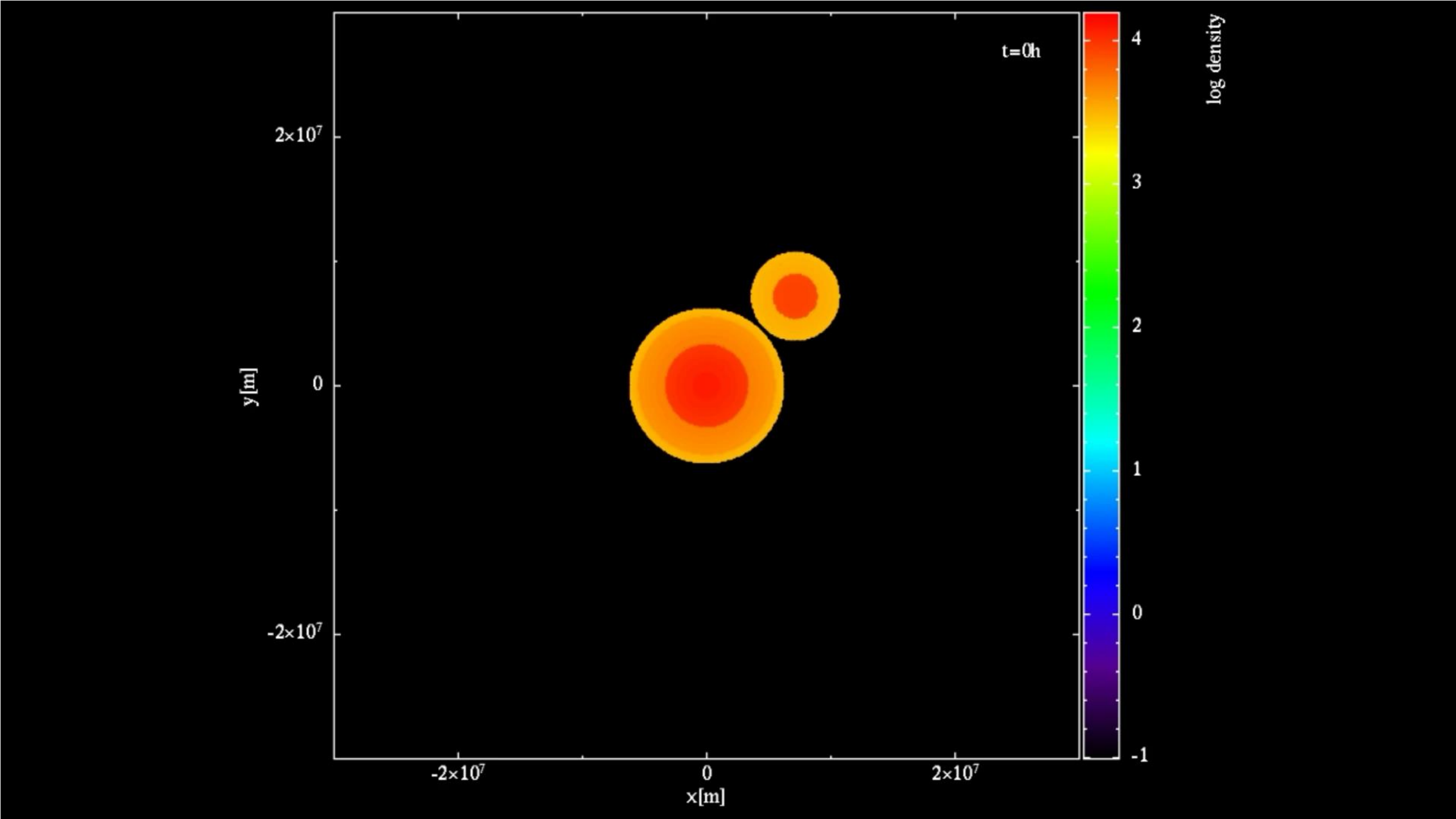


Smoothed Particle Hydrodynamics (SPH)

Or: "How I learned to stop worrying and love the Lagrangian"

Daniel Price




t=0h
2×10^7
0
-2×10^7
y[m]
-2×10^7
0
2×10^7
x[m]
4
3
2
1
0
-1
log density

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Discretizing a continous field

Continous scalar field:

$$f(r) = \int_V f(r')\delta(r - r')dV'$$

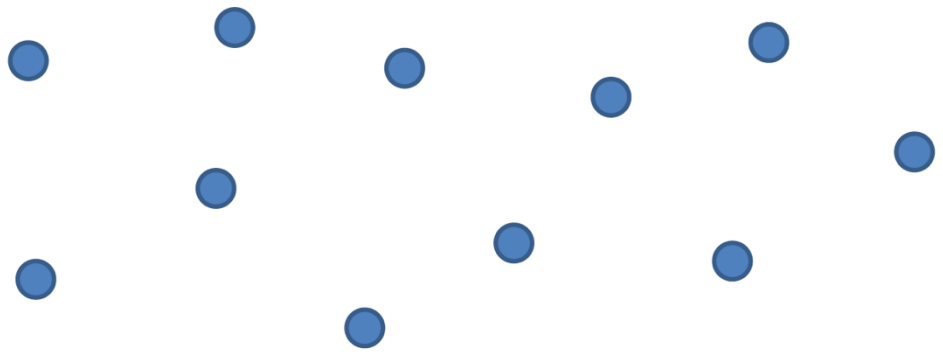
Discretizing a continous field

Continous scalar field:

$$f(r) = \int_V f(r')\delta(r - r')dV'$$

Introduce finite fluid elements (integral -> finite sum):

$$dV' = \frac{m_j}{\rho_j}. \qquad \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \, \delta(r_i - r_j)$$



Discretizing a continous field

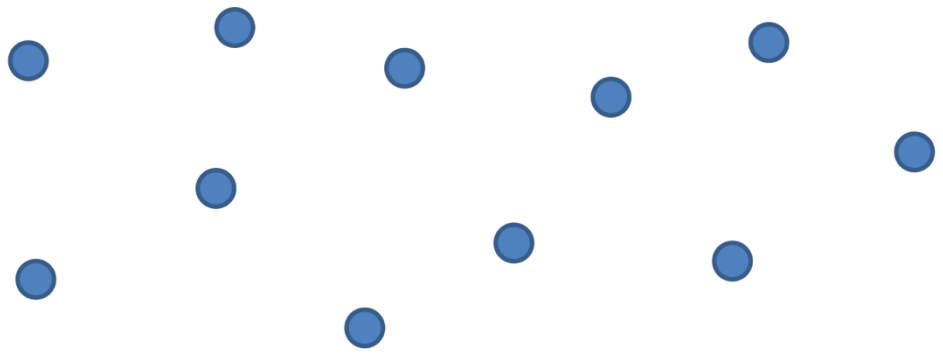
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Information given at points, however we need information everywhere to describe the continous field!

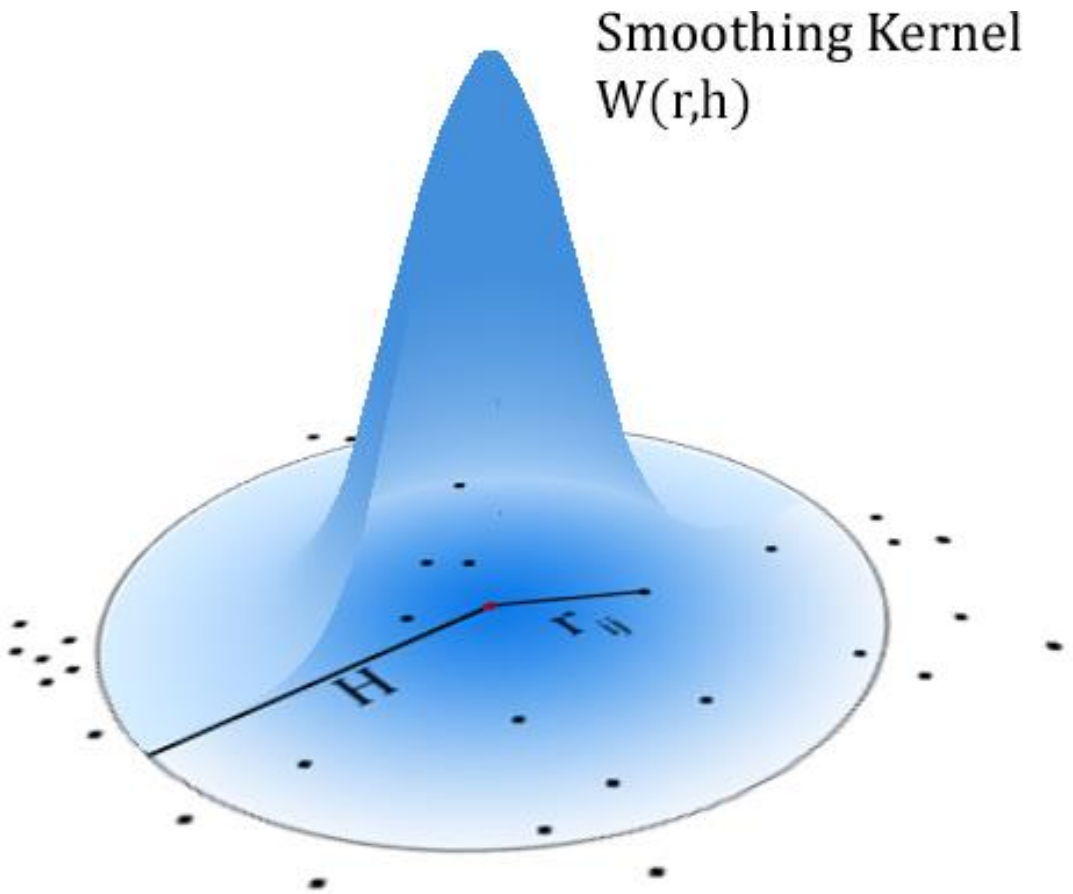


Smoothing Kernel

h represents our resolution scale similar to Δx in grid

- Smoothing Kernel:

1. $\lim_{h \to 0} W(r - r', h) = \delta(r - r')$
2. $\int_V W(r - r', h)\, dV' = 1$
3. Spherically symmetric
4. Smooth derivatives



Smoothing Kernel

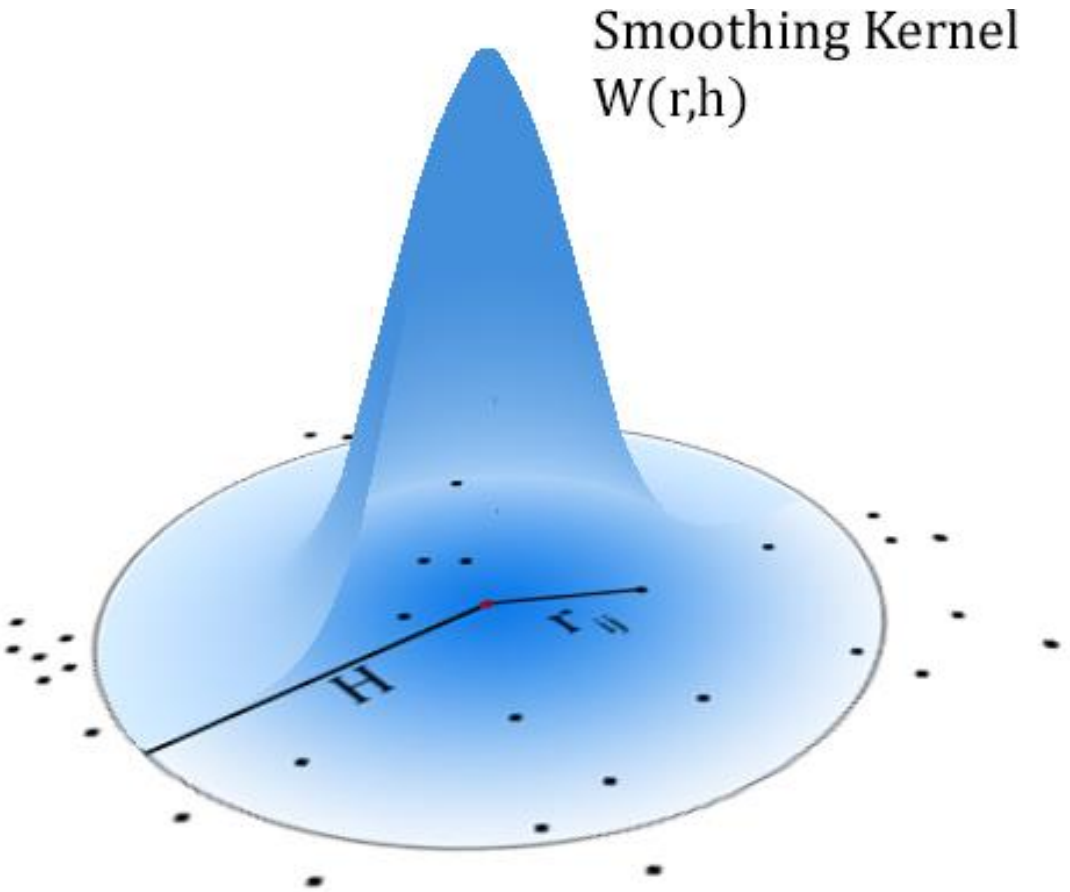
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- Gaussian function

Computational cost goes as $O(N^2)$



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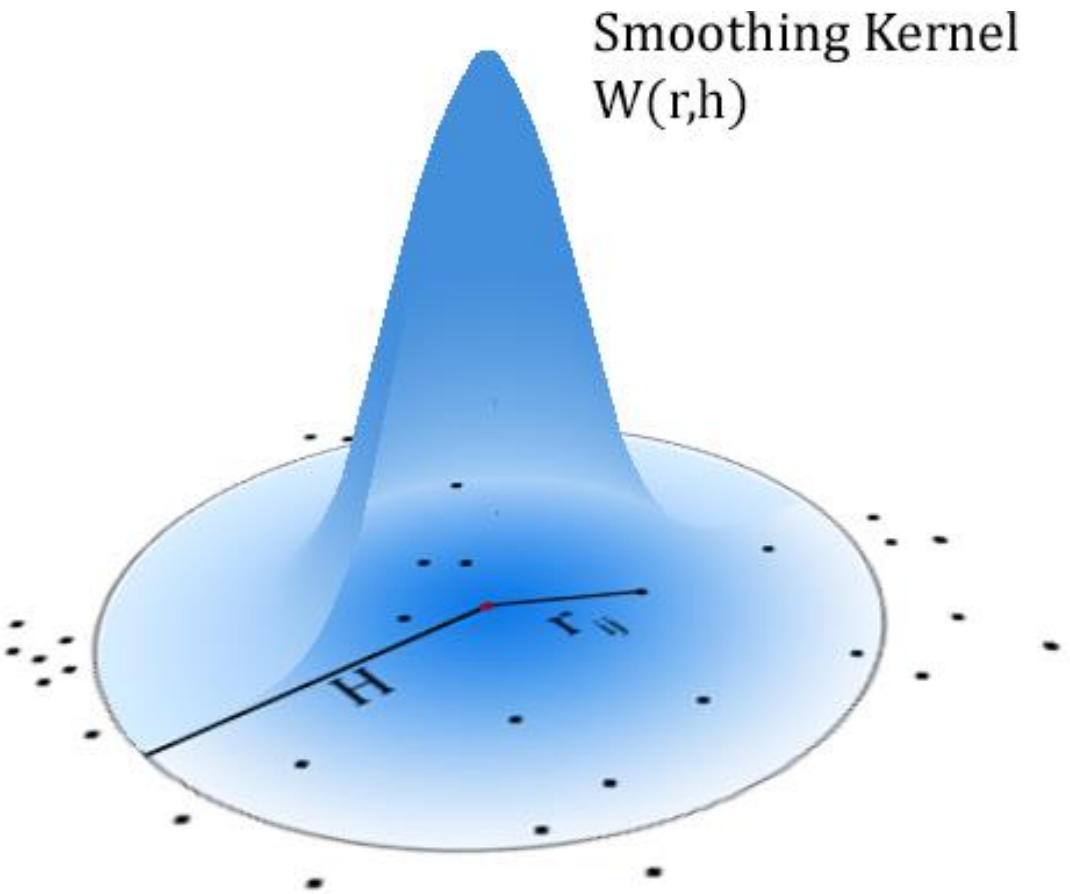
- Gaussian function

Computational cost goes as $O(N^2)$

- Compact support

Computational cost goes as $O(NN_{neigh})$

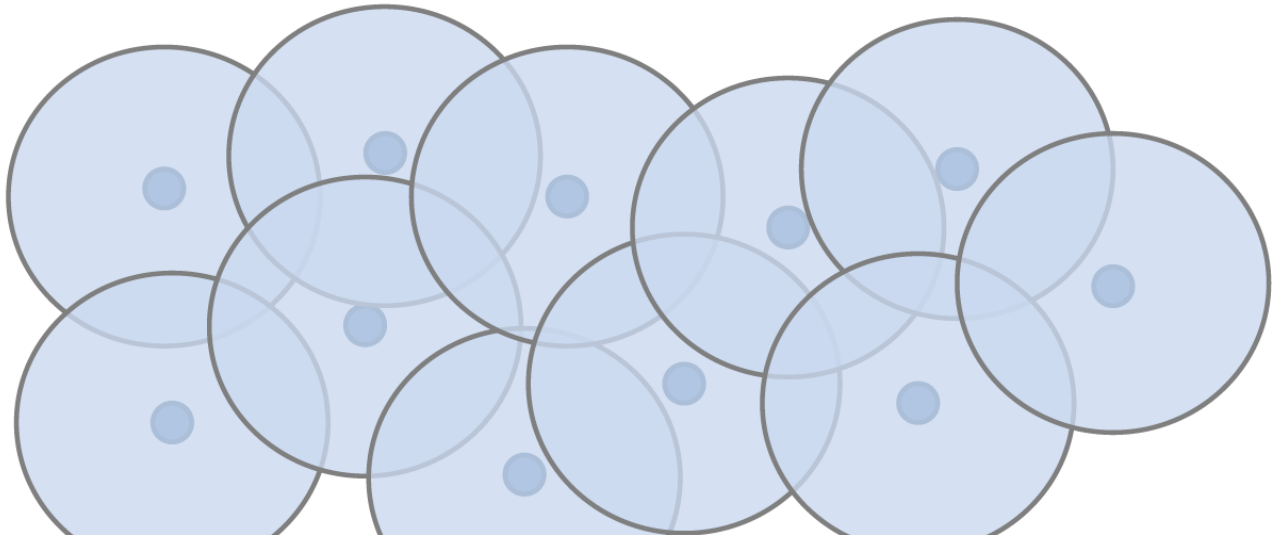
$W(r, h) = 0\ when\ r > H$



Discretizing a continous field

- Kernel interpolation:

$$f(r) = \int_V f(r')W(r - r', h)dV' + O(h^2)$$



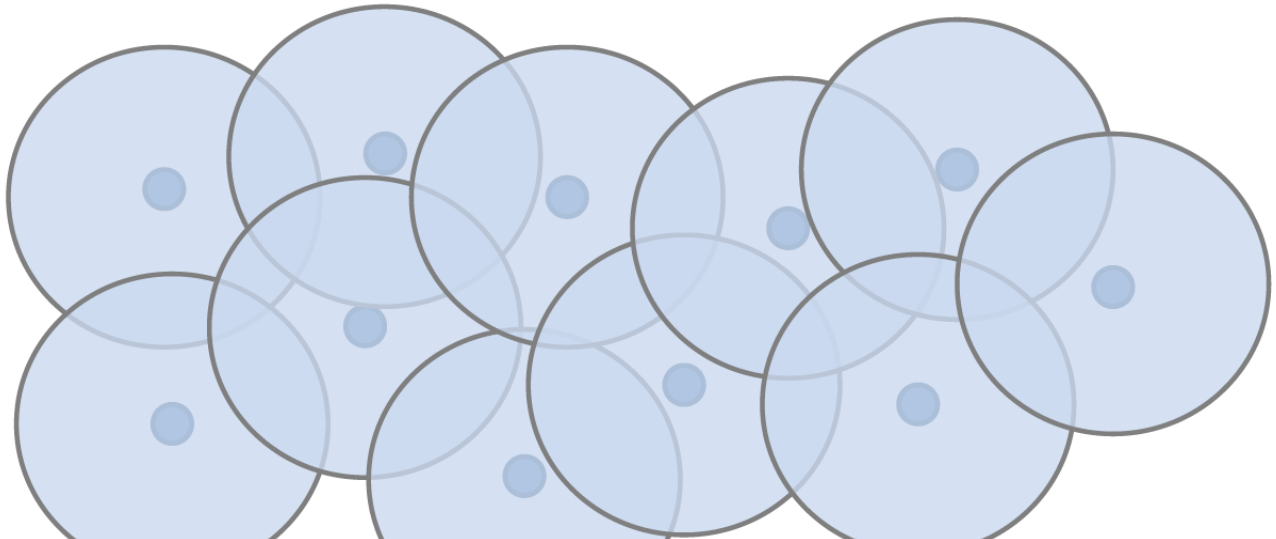
Discretizing a continous field

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- Discretize the continous field:

$$f_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \, W(r_i - r_j, h)$$



SPH derivatives

$$f_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \, W(r_i - r_j, h)$$

SPH derivatives

$$f_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \, W(r_i - r_j, h)$$

Take derivatives of the approximate form

$$\nabla f_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \nabla \, W(r_i - r_j, h)$$

$$\nabla \cdot \boldsymbol{f}_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} \boldsymbol{f}_j \cdot \nabla \, W(r_i - r_j, h)$$

SPH derivatives

The continuity equation:

$$\frac{d\rho}{dt} = -\rho \, \nabla \cdot \boldsymbol{v}$$

Discretize as in previous slide:

$$\frac{d\rho_i}{dt} = -\sum_j \frac{\mathrm{m_j}}{\rho_j} \boldsymbol{v}_j \cdot \boldsymbol{\nabla} W(r_i - r_j, h)$$

Does not ensure zero contribution at constant relative velocity.

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SPH derivatives

The continuity equation:

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Discretize as in previous slide:

$$\frac{d\rho_i}{dt} = -\sum_j \frac{\mathrm{m_j}}{\rho_j} \boldsymbol{v}_j \cdot \boldsymbol{\nabla} W(r_i - r_j, h)$$

Does not ensure zero contribution at constant relative velocity.

SPH derivatives

New gradient operator

$$\nabla f \approx \frac{1}{\rho}[\nabla(\rho f) - f\nabla\rho]$$

$$= \frac{1}{\rho_i}\sum_j \mathrm{m_j}(f_j - f_i)\boldsymbol{\nabla} W(r_i - r_j, h)$$

Zero for constant functions $\boldsymbol{f}$

We then get:

$$\frac{d\rho_i}{dt} = \sum_j m_j(\boldsymbol{v}_i - \boldsymbol{v_j}) \cdot \boldsymbol{\nabla} W(r_i - r_j, h)$$

SPH derivatives

$$f_i \approx \sum_j \frac{\mathrm{m_j}}{\rho_j} f_j \, W(r_i - r_j, h)$$

SPH derivatives

Interpolate density

$$\rho_i = \sum_j \mathrm{m_j} W(r - r_j, h)$$

Time derivative of density estimate (constant h):

$$\frac{d\rho_i}{dt} = \sum_j m_j(\boldsymbol{v}_i - \boldsymbol{v_j}) \cdot \boldsymbol{\nabla} W(r_i - r_j, h) = -\rho_i \, (\nabla \cdot \boldsymbol{v} \,)_i$$

Density estimate represent an exact solution to the continuity equation

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$$\rho_i = \sum_j \mathrm{m_j} W(r - r_j, h)$$

Equation of motion:

$$\frac{dv}{dt} = -\frac{\nabla P}{\rho}$$

?

Conservation of energy:

$$\frac{du}{dt} = -\frac{P}{\rho} \nabla \cdot v$$

$$\frac{du_i}{dt} = \frac{P_i}{\rho_i^2} \sum_j m_j (\boldsymbol{v}_i - \boldsymbol{v_j}) \cdot \boldsymbol{\nabla} W(r_i - r_j, h)$$

The equations of motion in SPH

$$\frac{\mathrm{dv}}{\mathrm{dt}} = -\frac{\nabla \mathrm{P}}{\rho}$$

Gradients discretized as before

$$\left(\frac{dv_i}{dt}\right)_{hydro} = -\frac{1}{\rho_i^2}\sum_j m_j(P_j - P_i)\nabla W_{ij}(h)$$

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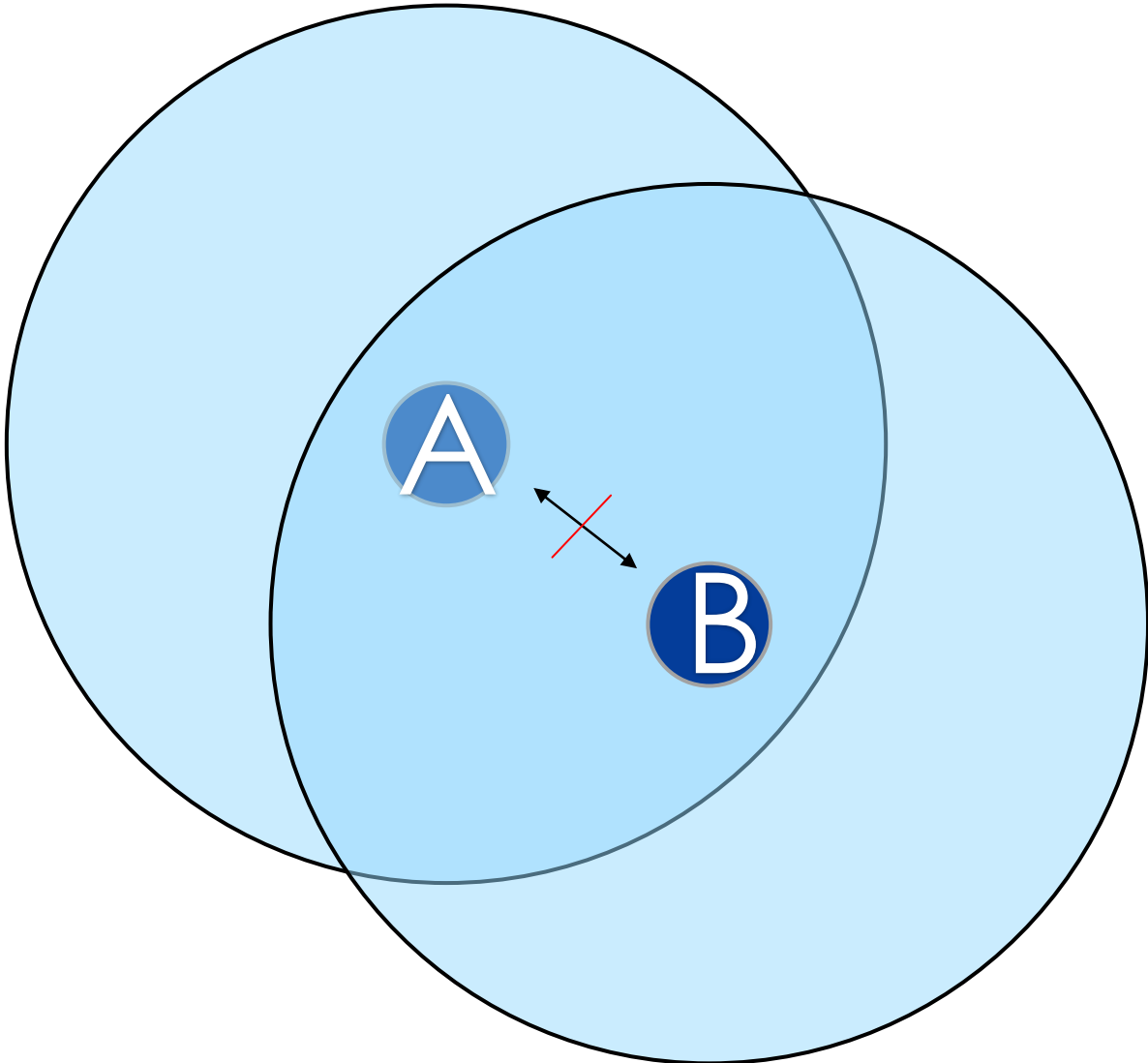
Problem: Not symmetric!

$$m_A\frac{dv_A}{dt} = -m_A m_B\frac{P_{BA}}{\rho_A^2}\nabla \; W_{AB}(h_a)$$

$$m_B\frac{dv_B}{dt} = m_A m_B\frac{P_{AB}}{\rho_B^2}\nabla \; W_{AB}(h_b)$$

$$\frac{dv_A}{dt} \neq -\frac{dv_B}{dt}$$

Does not conserve momentum!



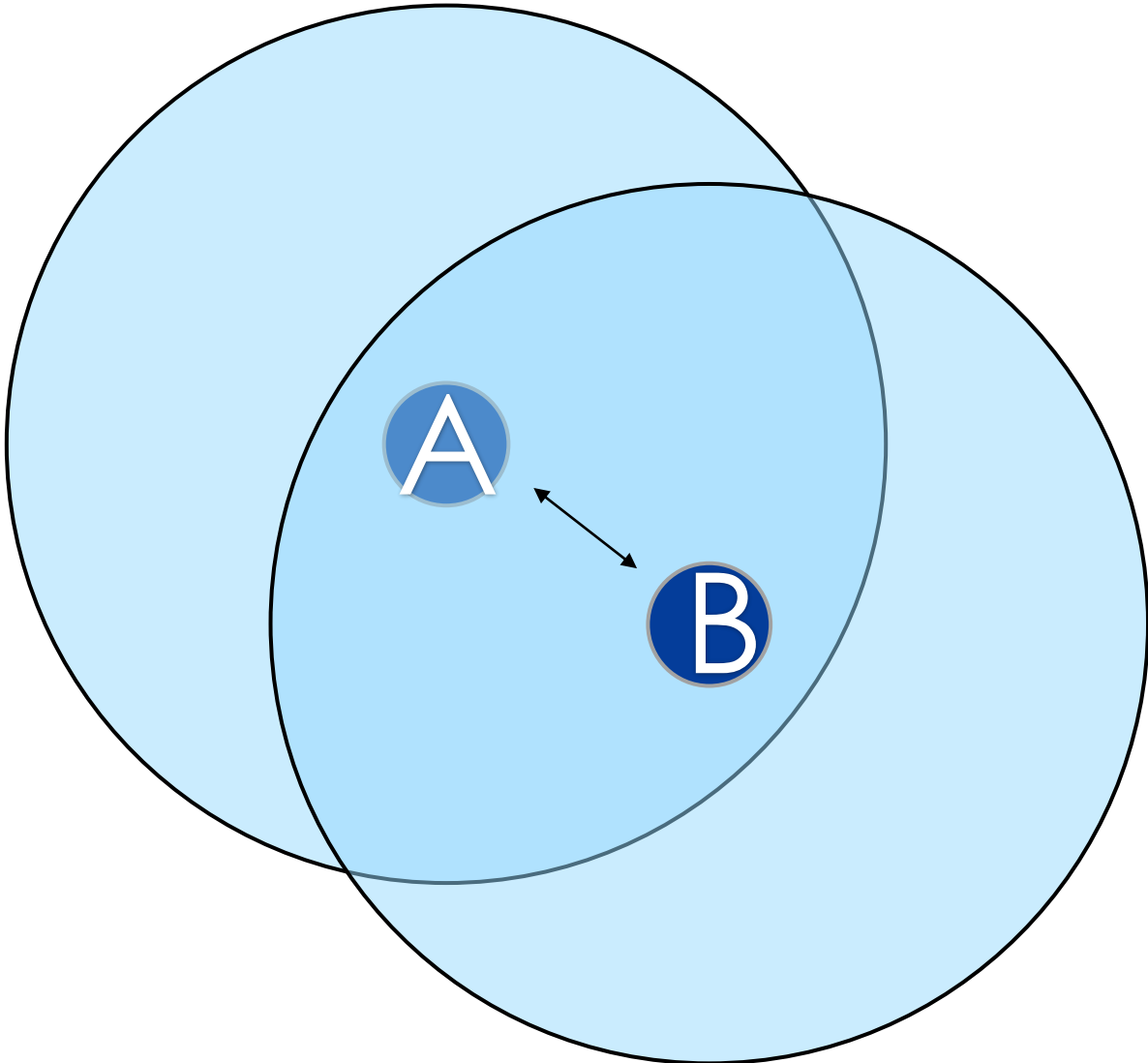
The equations of motion in SPH

$$\frac{\mathrm{dv}}{\mathrm{dt}} = -\frac{\nabla \mathrm{P}}{\rho}$$

One way to symmetrize:

$$\left(\frac{dv_i}{dt}\right)_{hydro} = -\sum_j m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2}\right) \nabla_i W_{ij}(h)$$

$$\frac{dv_A}{dt} = -\frac{dv_B}{dt}$$



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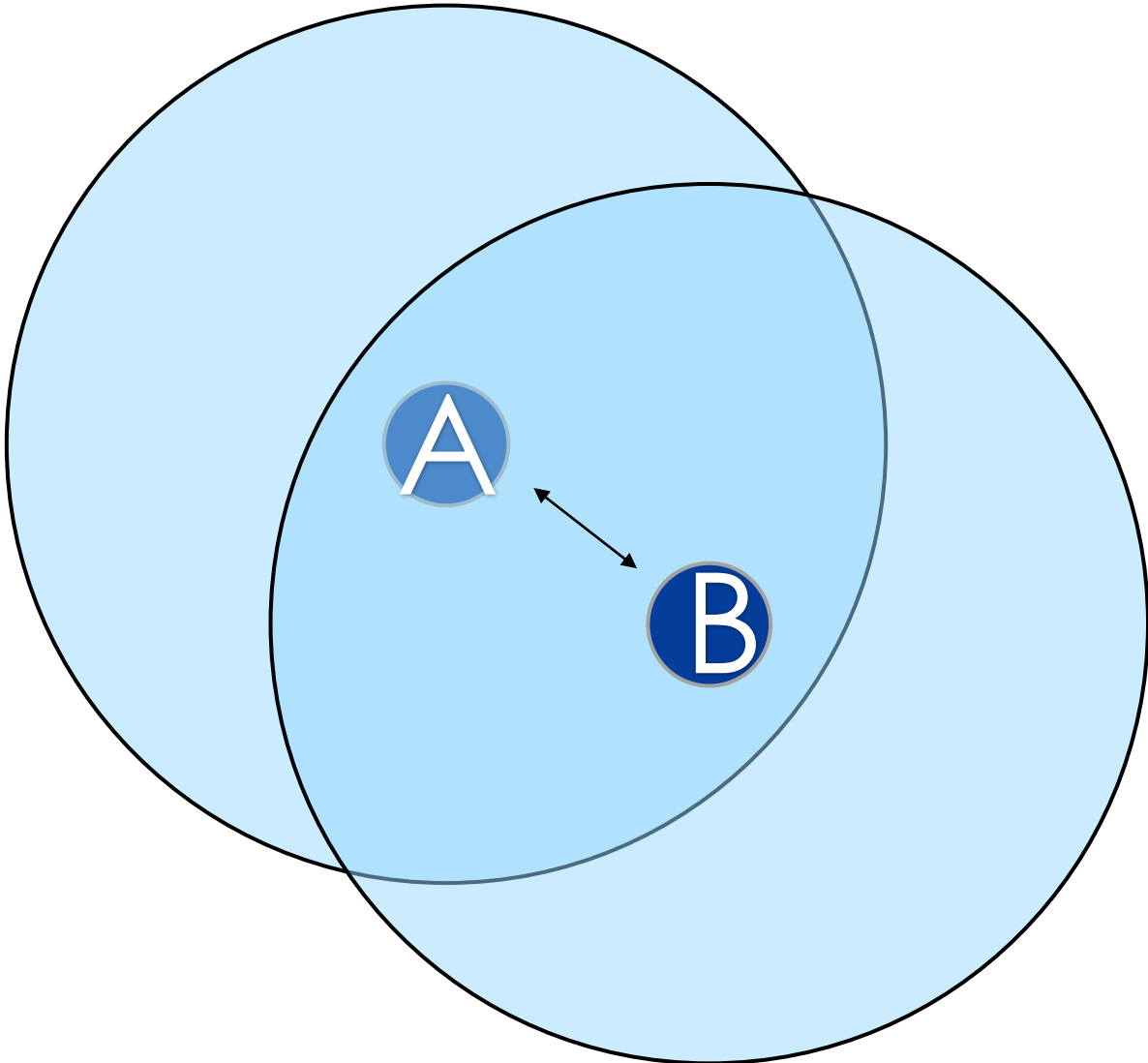
$$\left(\frac{dv_i}{dt}\right)_{hydro} = -\sum_j m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2}\right) \nabla_i W_{ij}(h)$$

$$\frac{dv_A}{dt} = -\frac{dv_B}{dt}$$

New gradient operator:

$$\nabla f \approx \rho \left[\frac{f}{\rho^2}\nabla(\rho) + \nabla\left(\frac{f}{\rho}\right)\right]$$

$$= \rho_i \sum_j \mathrm{m_j} \left(\frac{f_i}{\rho_i^2} + \frac{f_j}{\rho_j^2}\right) \boldsymbol{\nabla_i} W_{ij}(h)$$



SPH equations

$$\rho_i = \sum_j \mathrm{m_j} W_{ij}(h)$$

$$\frac{dv_i}{dt} = -\sum_j m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} \right) \nabla_i W_{ij}(h) + F_{extra}$$

$$\frac{du_i}{dt} = \frac{P_i}{\rho_i^2} \sum_j m_j (\boldsymbol{v}_i - \boldsymbol{v_j}) \cdot \nabla_i W_{ij}(h) + Q_{extra}$$

+ An EOS, for example $\mathrm{P_i} = (\gamma - 1)\rho_i u_i$

These equations can be derived from the principle of least action

From the principle of least action

See Price 2012 for full derivation

- ✓ Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial v}\right) - \frac{\partial L}{\partial r} = 0$$

- ✓ Lagrangian

$$\mathrm{L} = \sum_j m_j \left(\frac{1}{2} v_j^2 - u_j\right)$$

- ✓ First law of thermodynamics (isentropic)

$$du = \frac{P}{\rho^2} d\rho$$

- ✓ Density

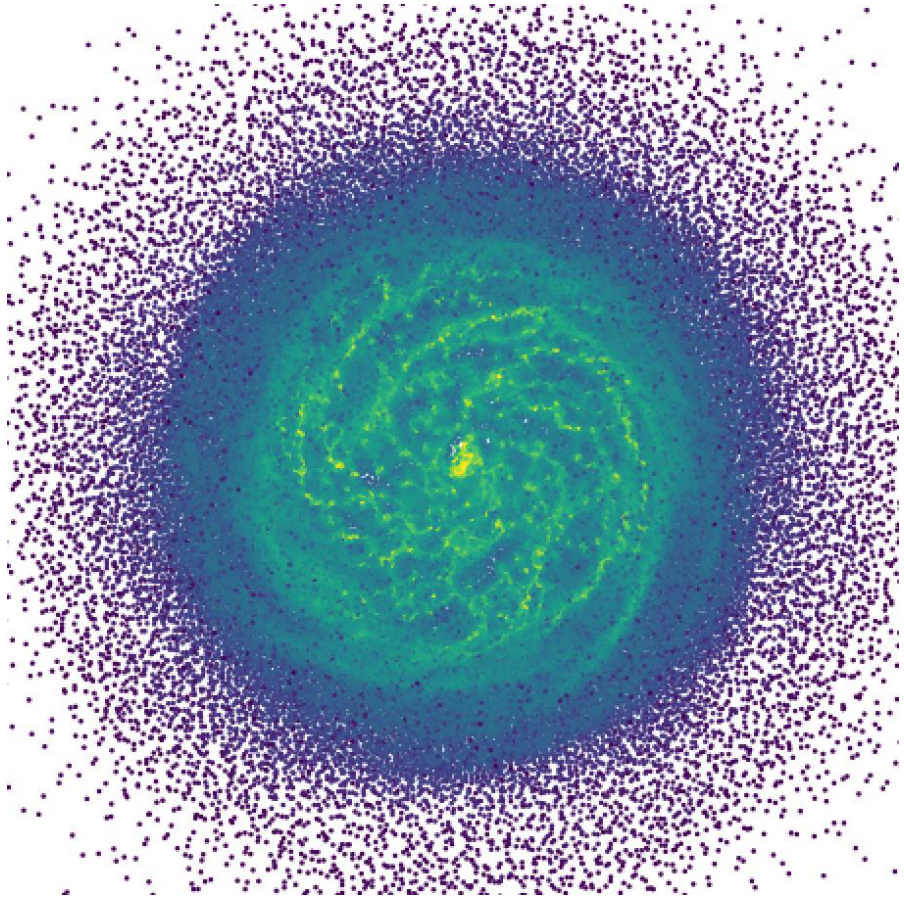
$$\rho_i = \sum_j m_j W_{ij}(h)$$

- ✓ Time derivative of density gives internal energy evolution

$$\frac{du_i}{dt} = \frac{P_i}{\rho_i^2} \frac{d\rho_i}{dt}$$

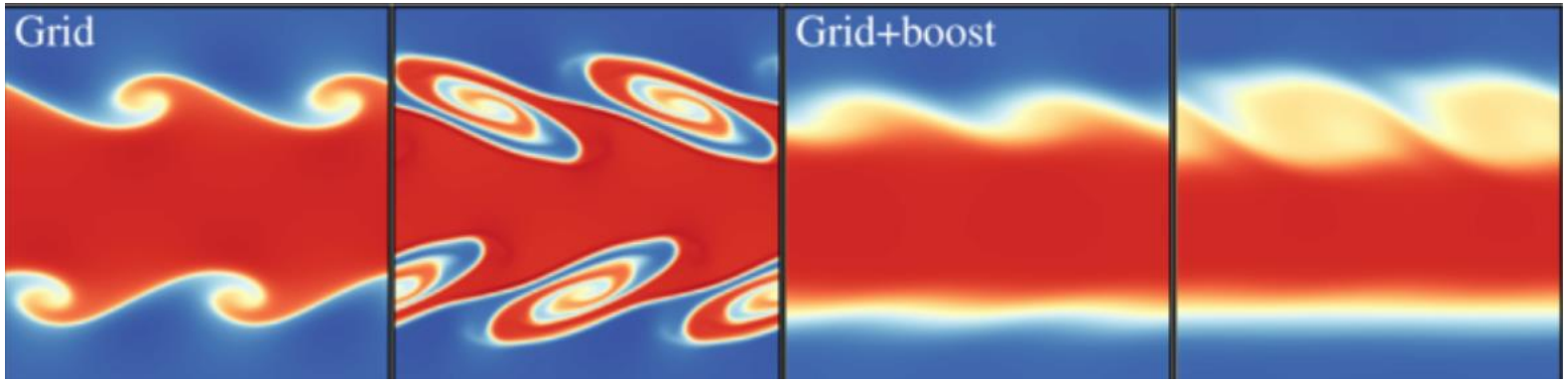
Advantages of SPH

- Resolution follows mass

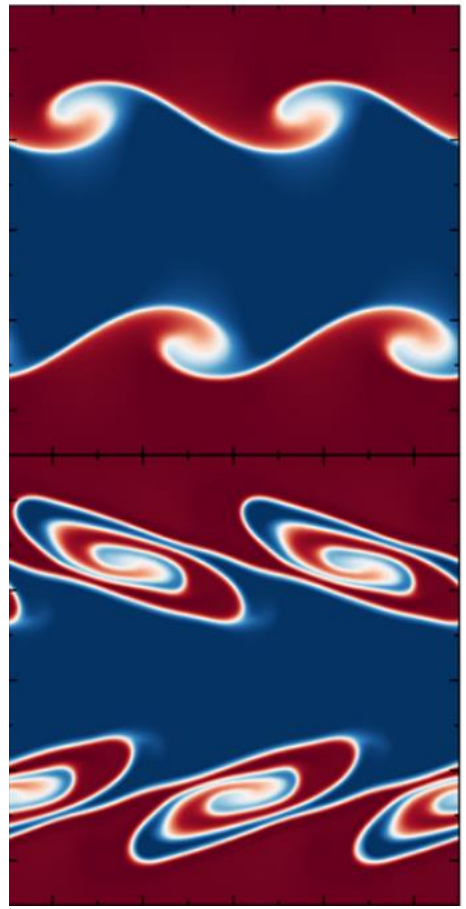


Advantages of SPH

- Resolution follows mass
- No intrinsic dissipation
- Galilean invariant



SPH



Advection/truncation errors in Eulerian codes are not Galilean invariant and increases with an imposed bulk velocity

Advantages of SPH

- Resolution follows mass
- No intrinsic dissipation
- Galilean invariant
- Exact solution to the continuity equation
- Excellent conservation of momentum, angular momentum, energy and entropy

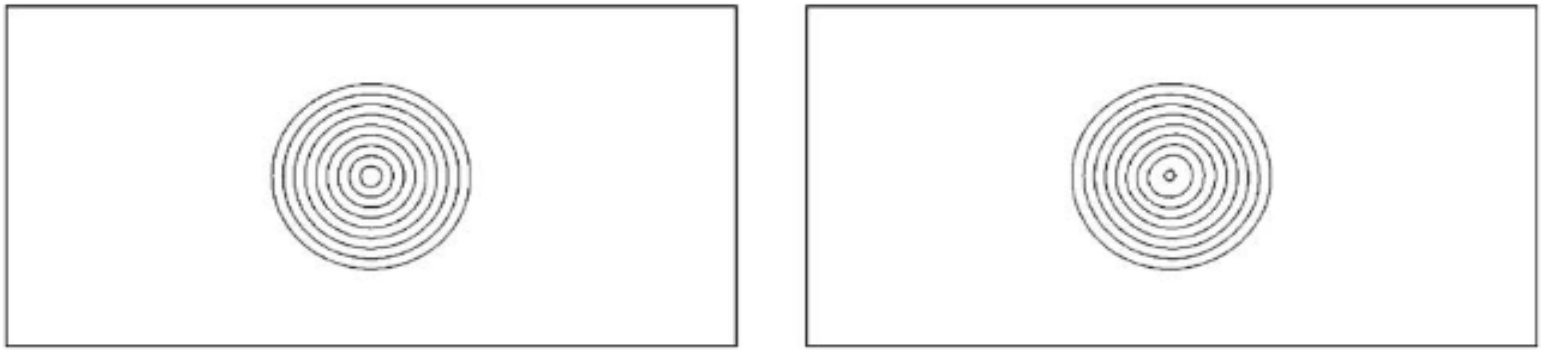
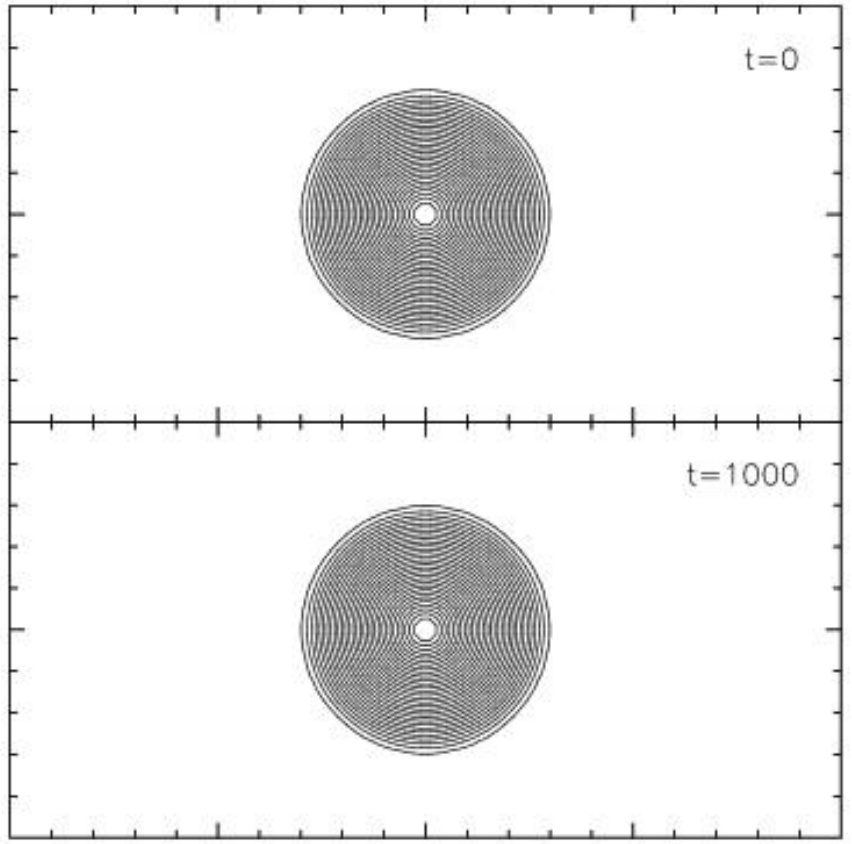


Fig. 8. Magnetic field lines at $t = 0$ (left) and $t = 2$ (right) using the CTU + CT integration algorithm.

GRID



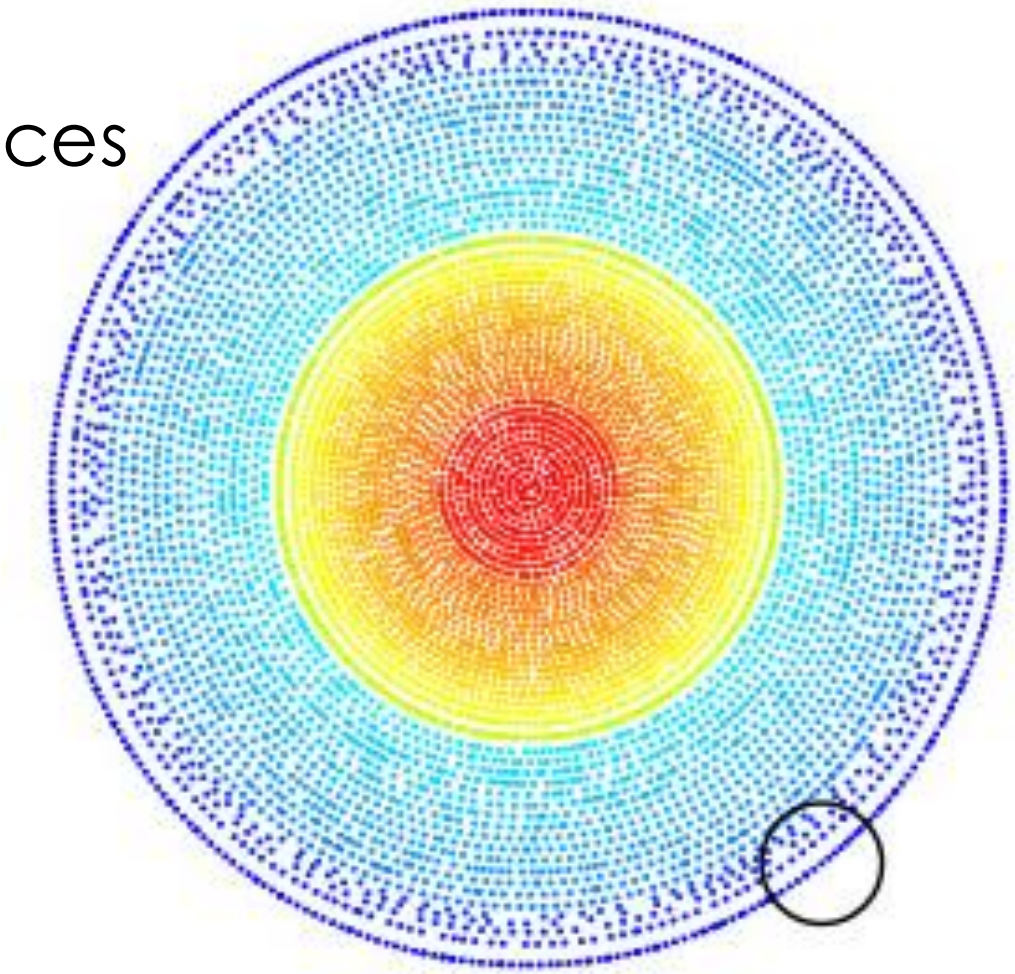
SPH

Advantages of SPH

- Resolution follows mass
- No intrinsic dissipation
- Galilean invariant
- Exact solution to the continuity equation
- Excellent conservation of momentum, angular momentum, energy and entropy
- Easy tracking of material evolution
- Couples directly with N-body codes (gravity)

Disadvantages of SPH

- Resolution follows mass
- Density discountinuities/free surfaces



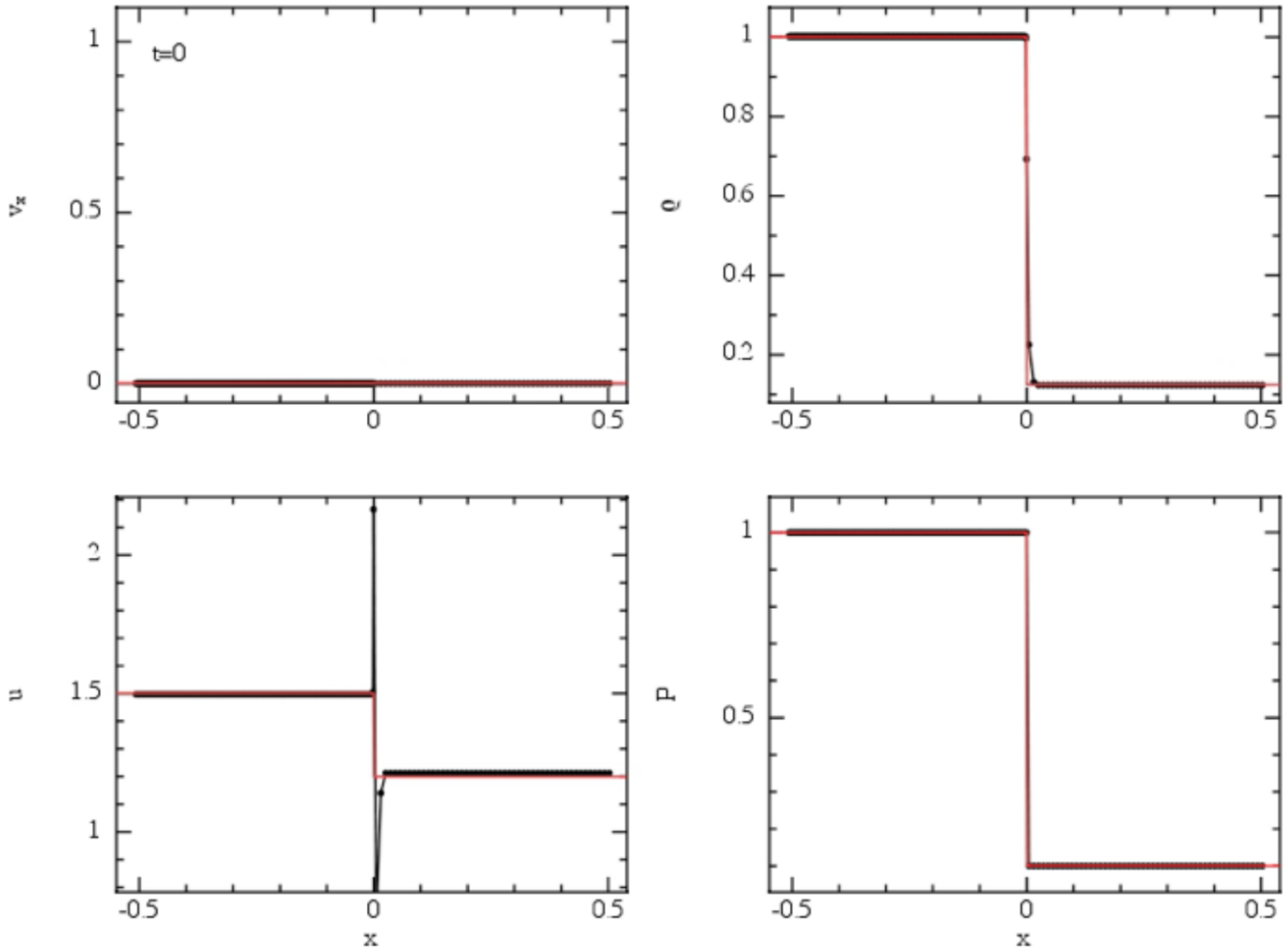
Disadvantages of SPH

- Resolution follows mass
- Density discountinuities/free surfaces
- Certain flux conserving prescriptions not available

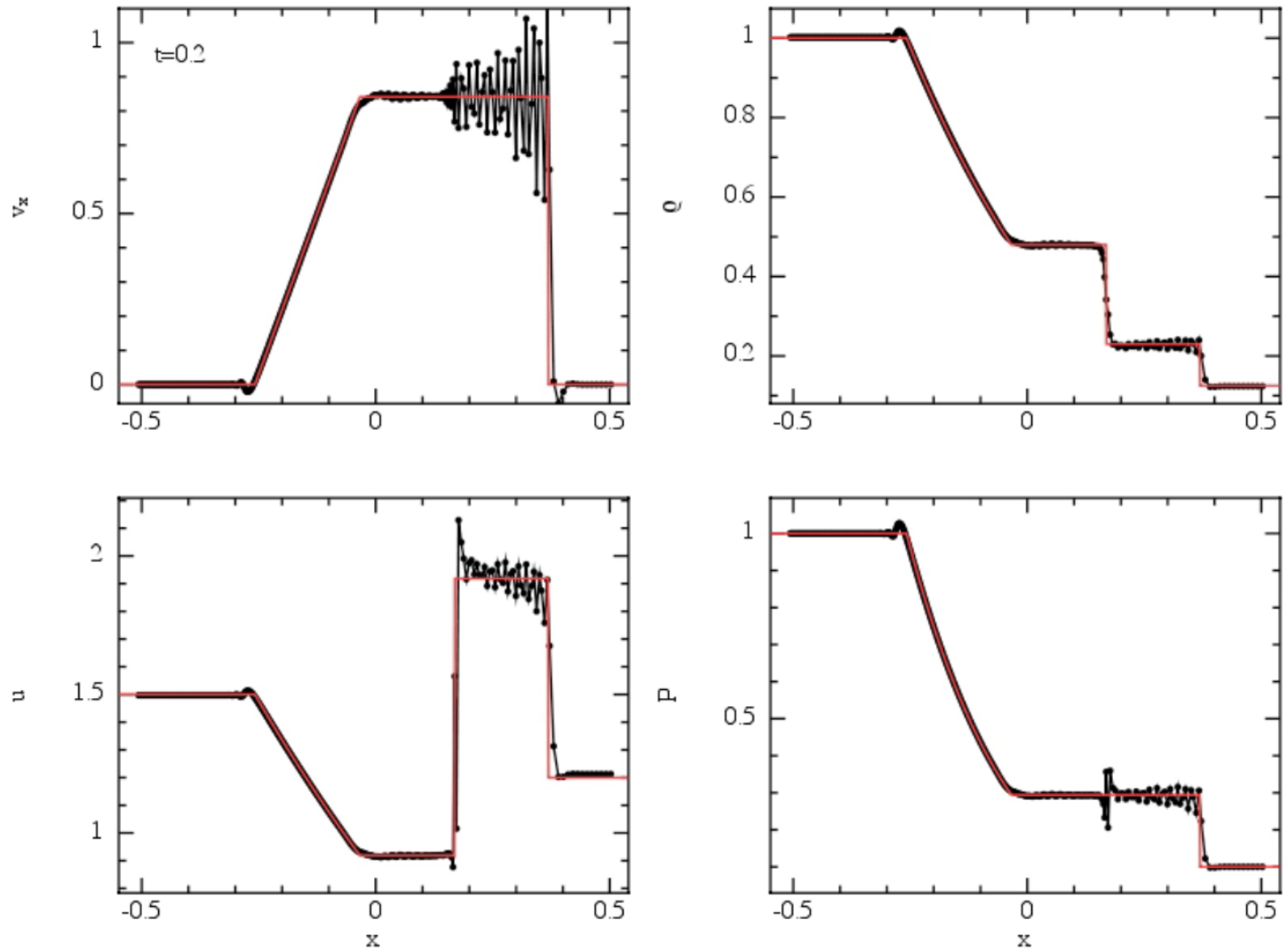
Disadvantages of SPH

- Resolution follows mass
- Density discountinuities/free surfaces
- Certain flux conserving prescriptions not available
- No intrinsic dissipation
- Dissipation terms needs to be explicitly added to smooth out discountinuities

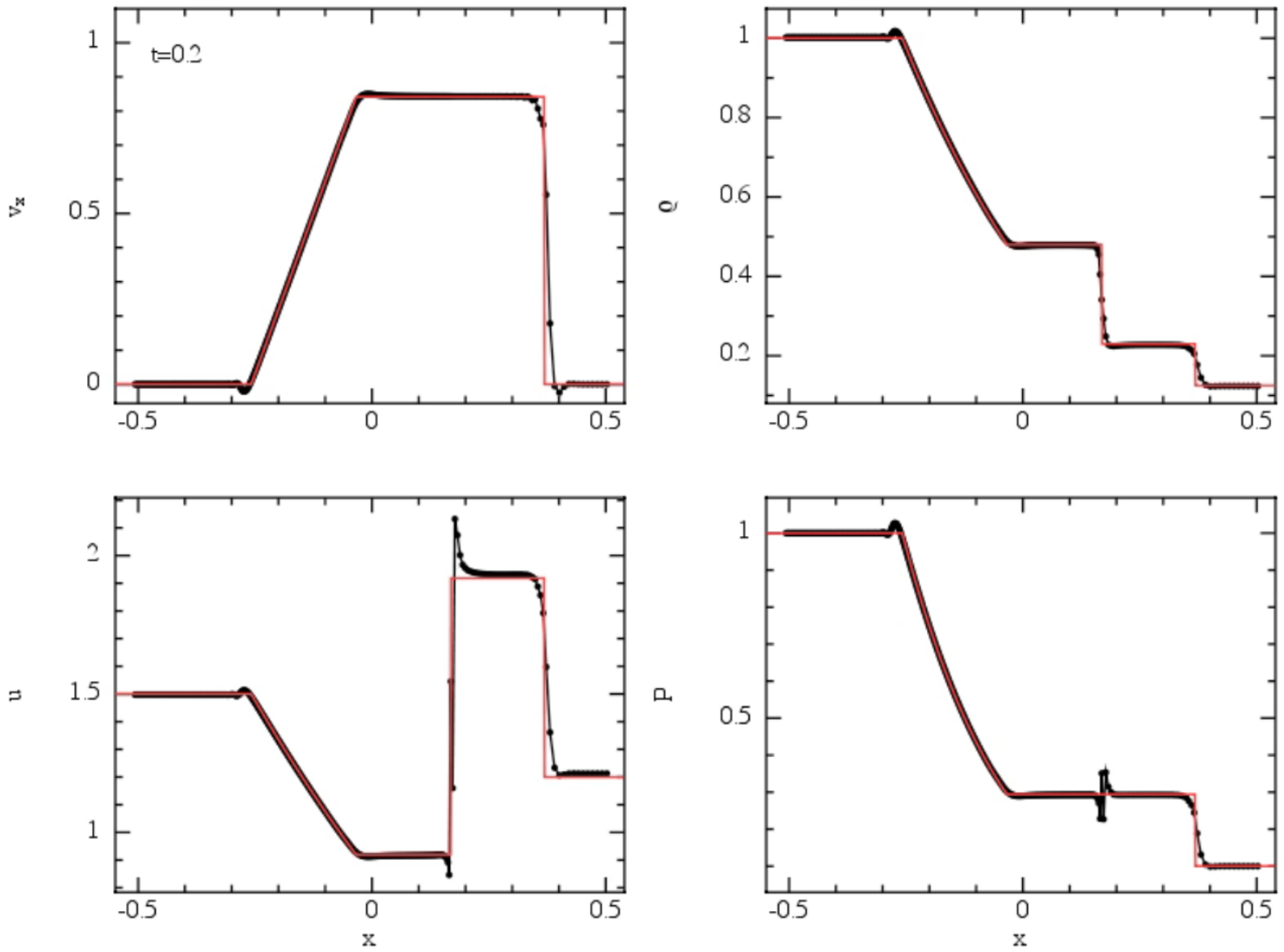
Sod Shock Tube



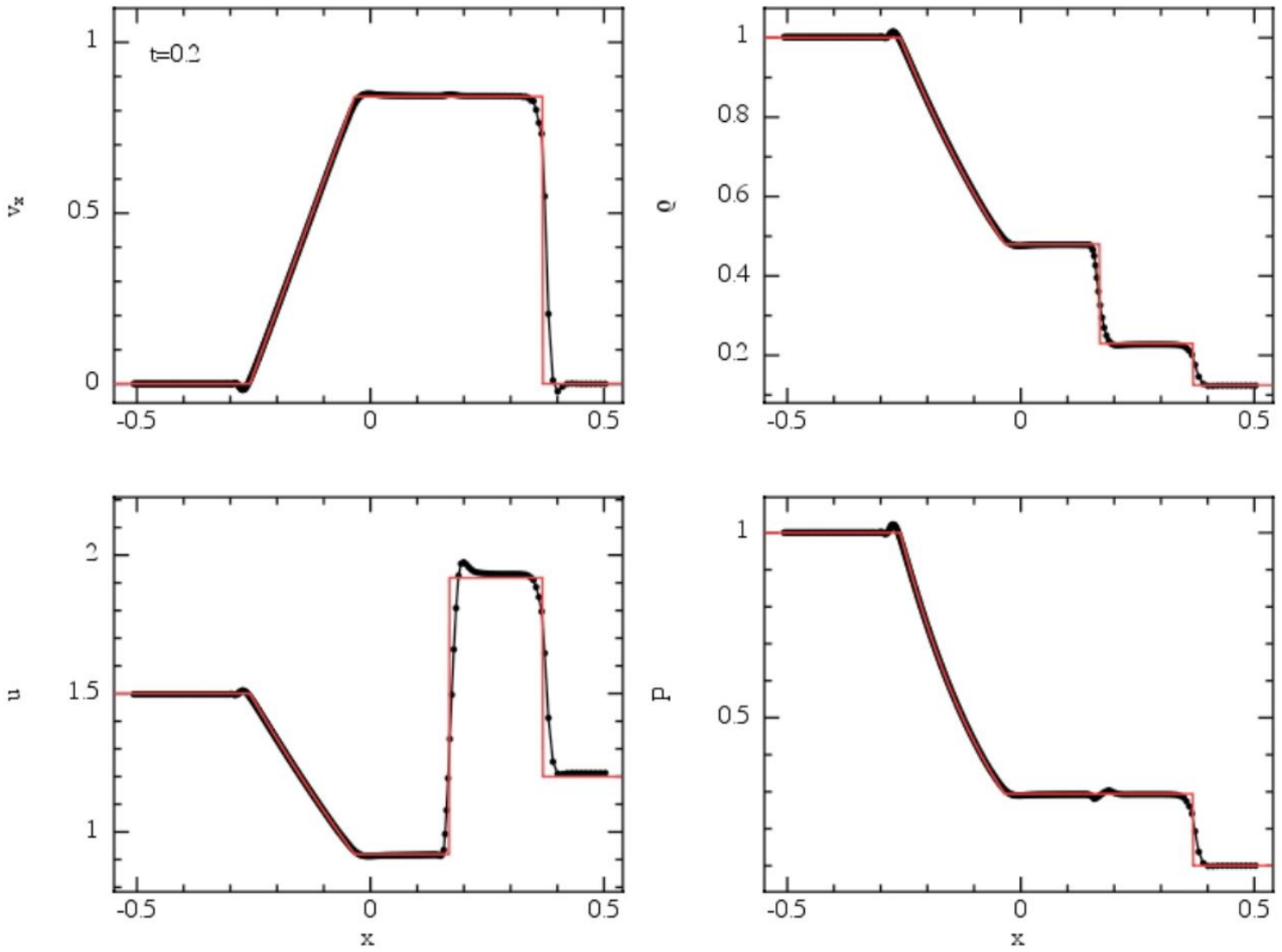
No Dissipation



Adding Viscosity



Adding Thermal Conductivity

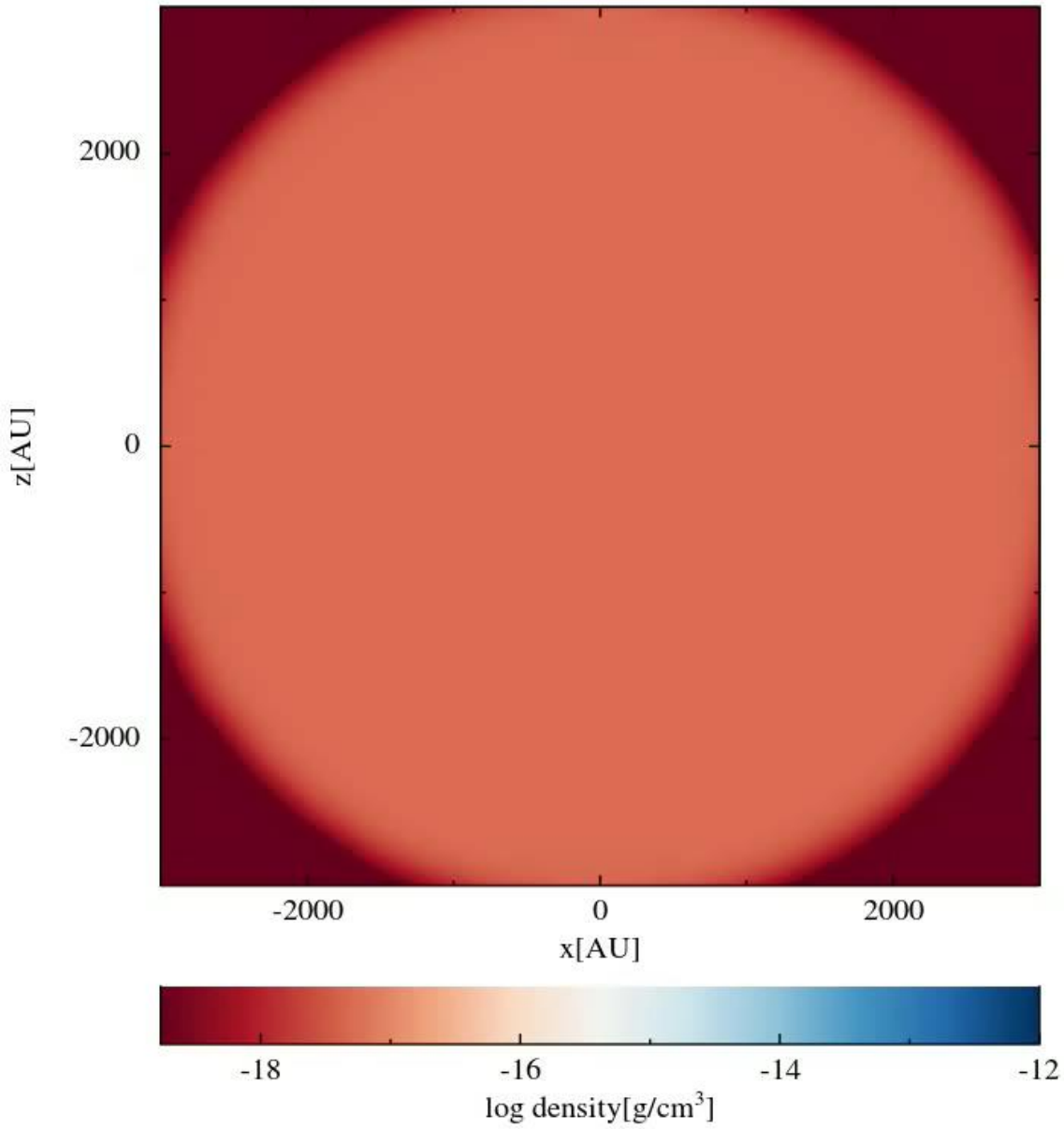


Disadvantages of SPH

- Resolution follows mass
- Density discountinuities/free surfaces
- Certain flux conserving prescriptions not available
- No intrinsic dissipation
- Dissipation terms needs to be explicitly added to smooth out discountinuities
- Errors depend on particle distribution within kernel

$$E_0 = \sum_j \frac{m_j}{\rho_j} \nabla W_{ij}(h) \rightarrow 0$$

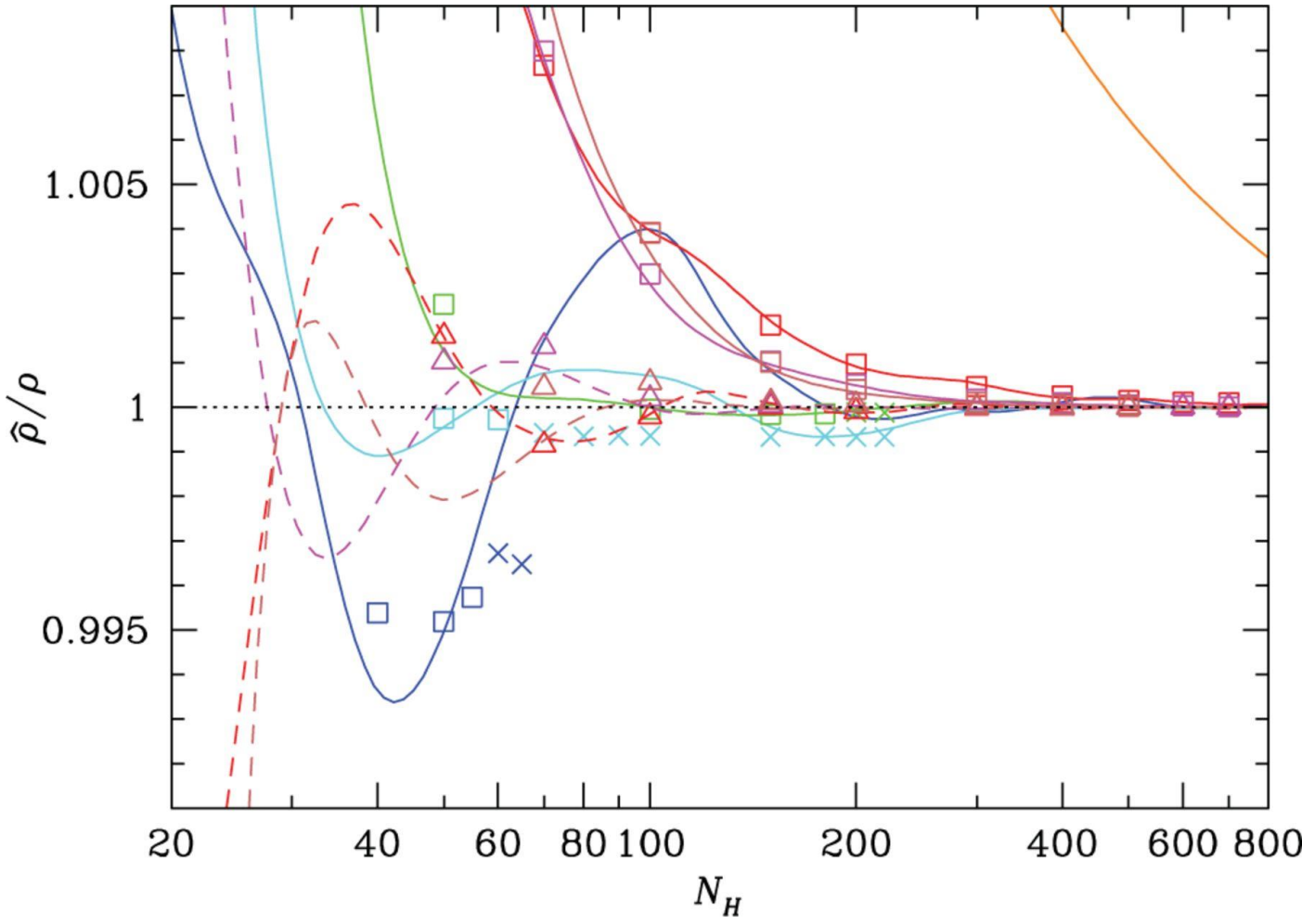
GDSPH 50^3. $\mu = 10$





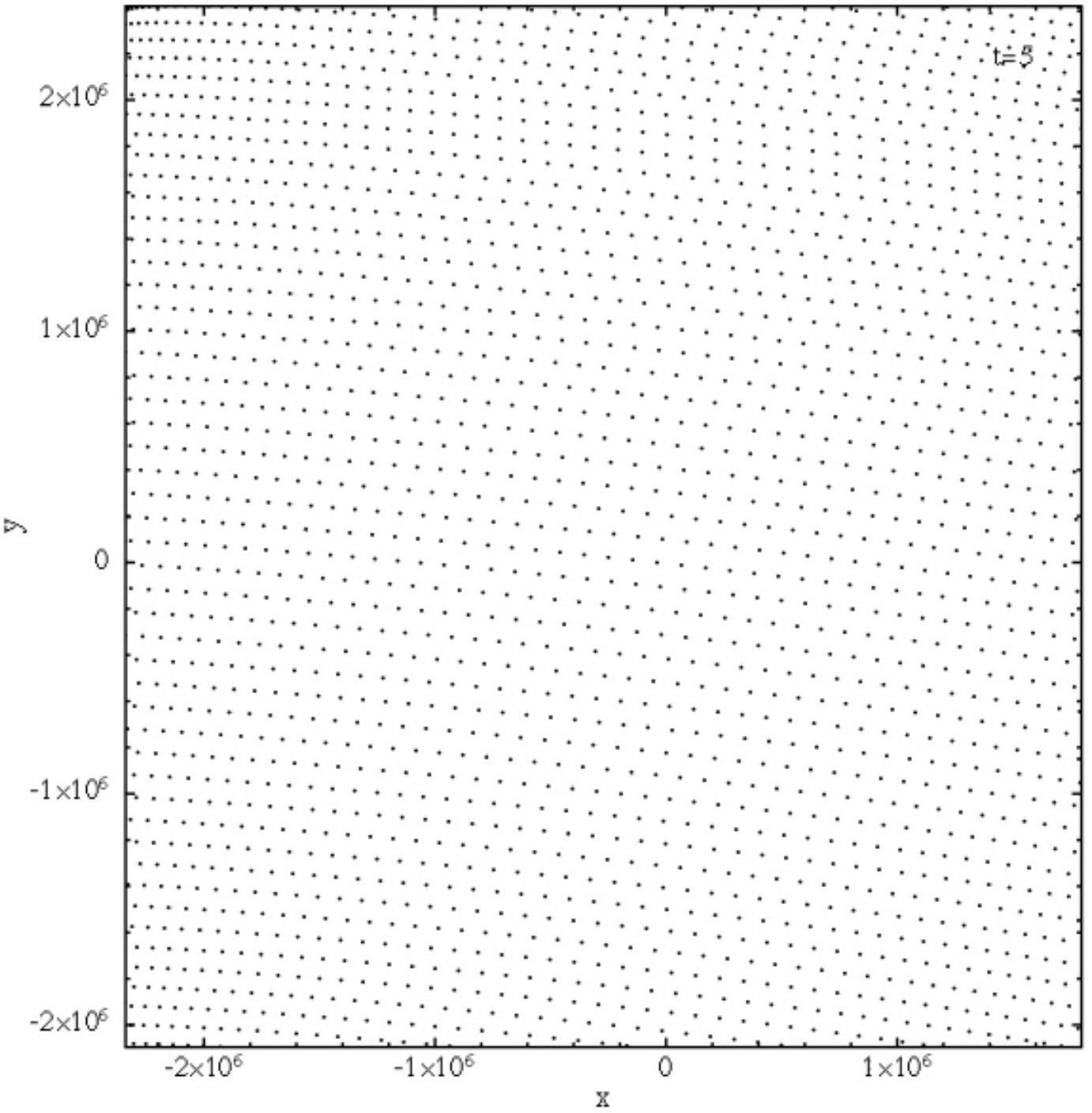
Resolution in SPH

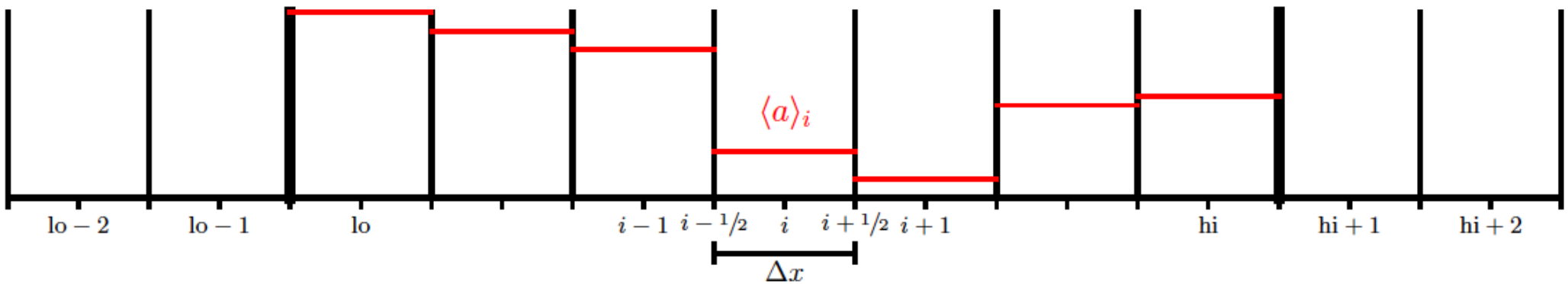
- Error from kernel interpolation $O(h^2)$
- Sampling error from discretization scales with N_{neigh}



Resolution in SPH

- Error from kernel interpolation $O(h^2)$
- Sampling error from discretization scales with N_{neigh}
- However, cannot simply increase N_{neigh} for most kernels




$\langle a \rangle_i$
lo − 2
lo − 1
lo
$i-1$
$i-1/2$
i
$i+1/2$
$i+1$
hi
hi + 1
hi + 2
Δx

Generalized conservative SPH form

- Exact conservation of energy and momentum requires:
  - **Antisymmetric:** $\nabla f \approx \frac{1}{\phi}[\nabla(\phi f) - f\nabla\phi] = \sum_j \frac{\mathrm{m_j}}{\rho_j}\frac{\phi_j}{\phi_i}\left(f_j - f_i\right)\boldsymbol{\nabla_i} W_{ij}(h)$
  - **Symmetric:** $\nabla f \approx \phi\left[\frac{f}{\phi^2}\nabla(\phi) + \nabla\left(\frac{f}{\phi}\right)\right] = \sum_j \frac{\mathrm{m_j}}{\rho_j}\left(\frac{\phi_j}{\phi_i}f_i + \frac{\phi_i}{\phi_j}f_j\right)\boldsymbol{\nabla_i} W_{ij}(h)$

Previous derivation $\phi = \rho$

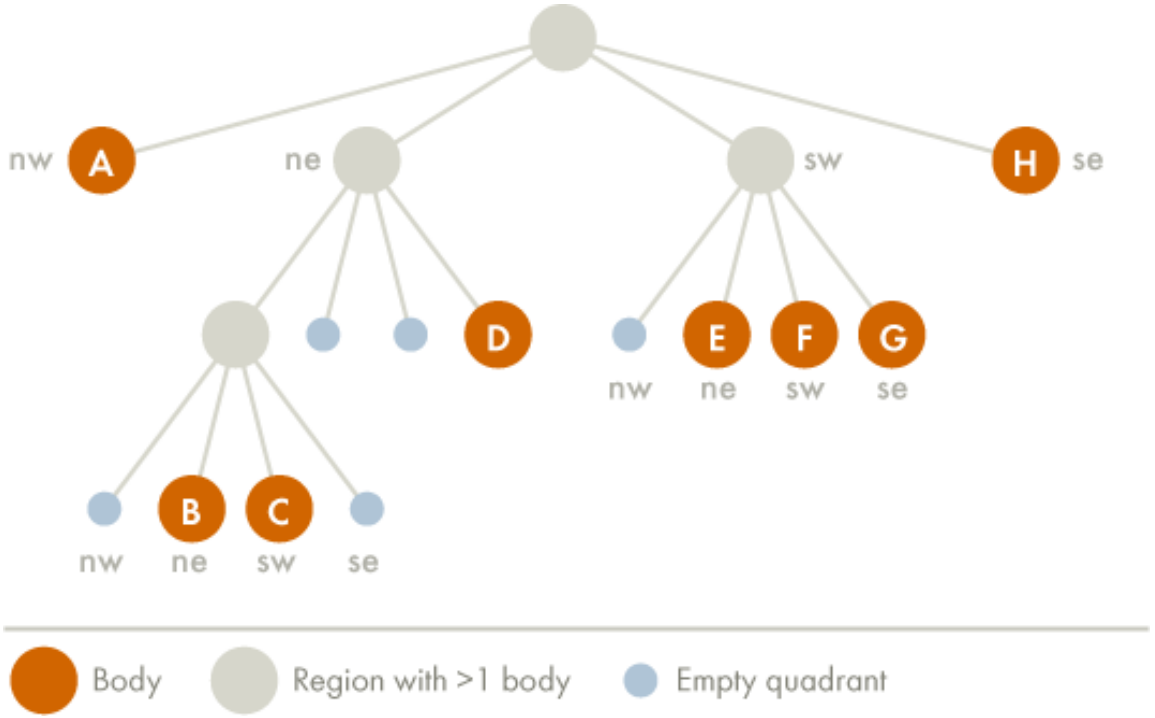
- However alternative formulations requires corrections to preserve entropy.
- Additional corrections are also required when variable smoothing lengths are used.

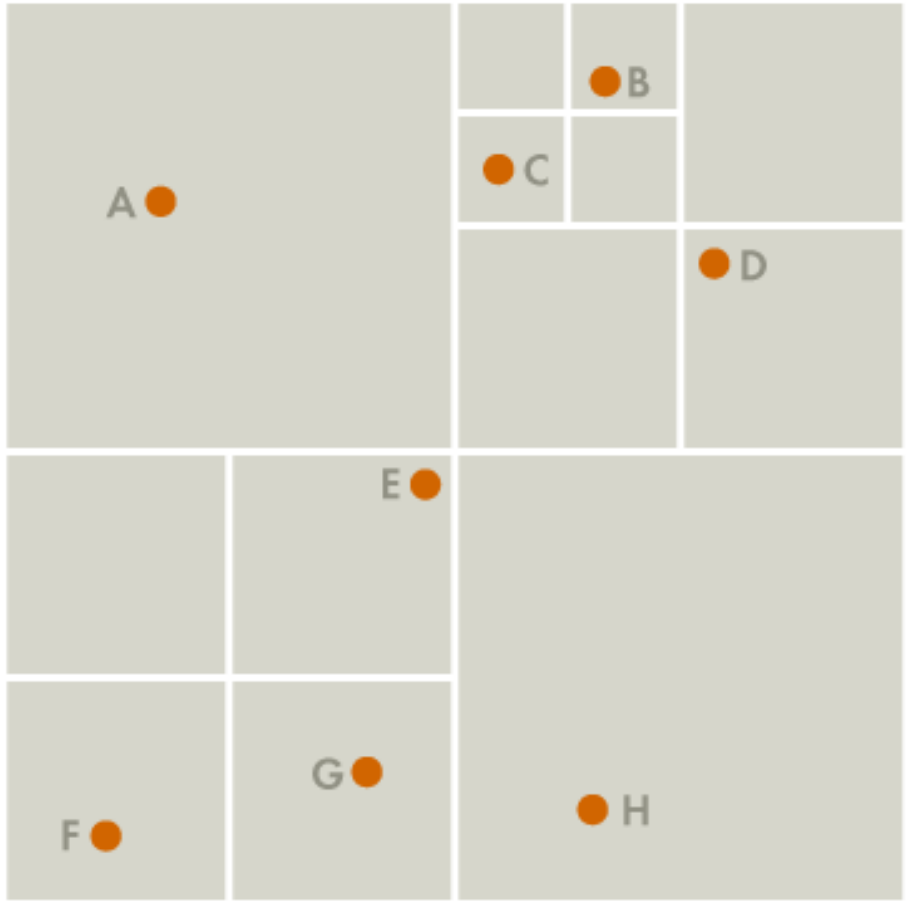
Optimization

- Cost of pure summation $O(N^2)$
  Highly accurate but expensive

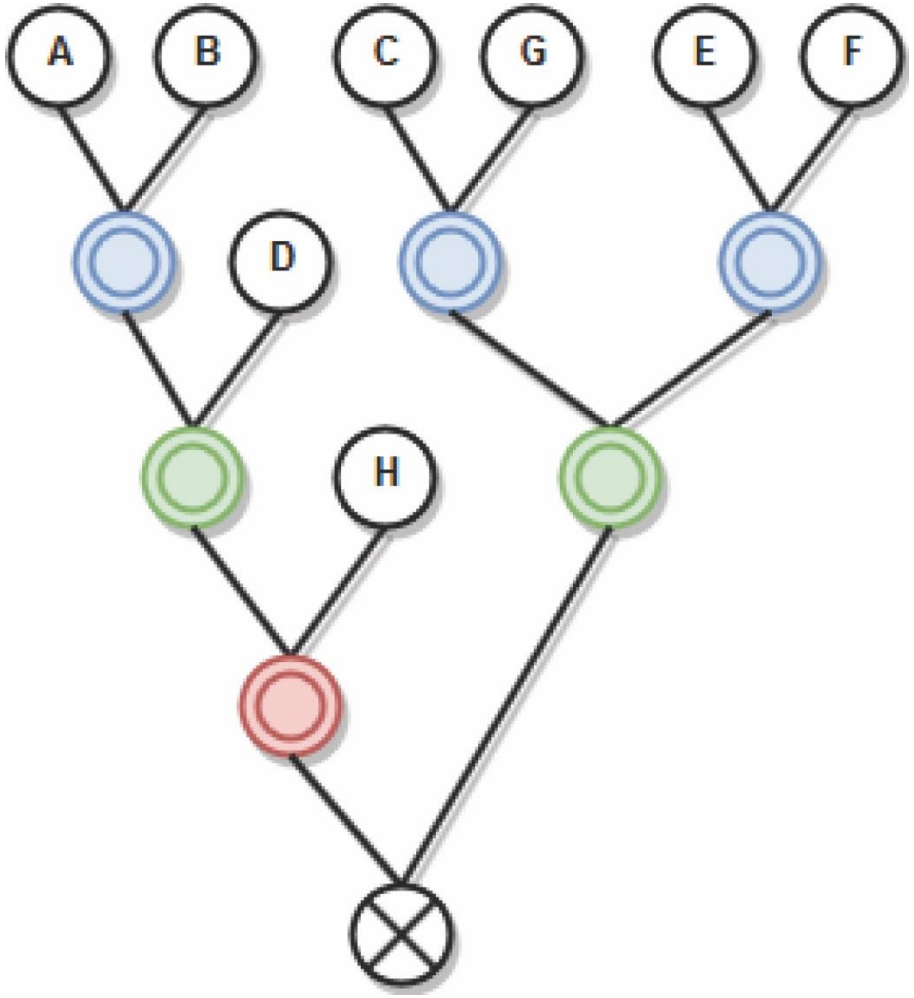
- Tree method reduces cost $O(N \log N)$
  Reduces cost at the expense of accuracy
- Particles grouped into nodes

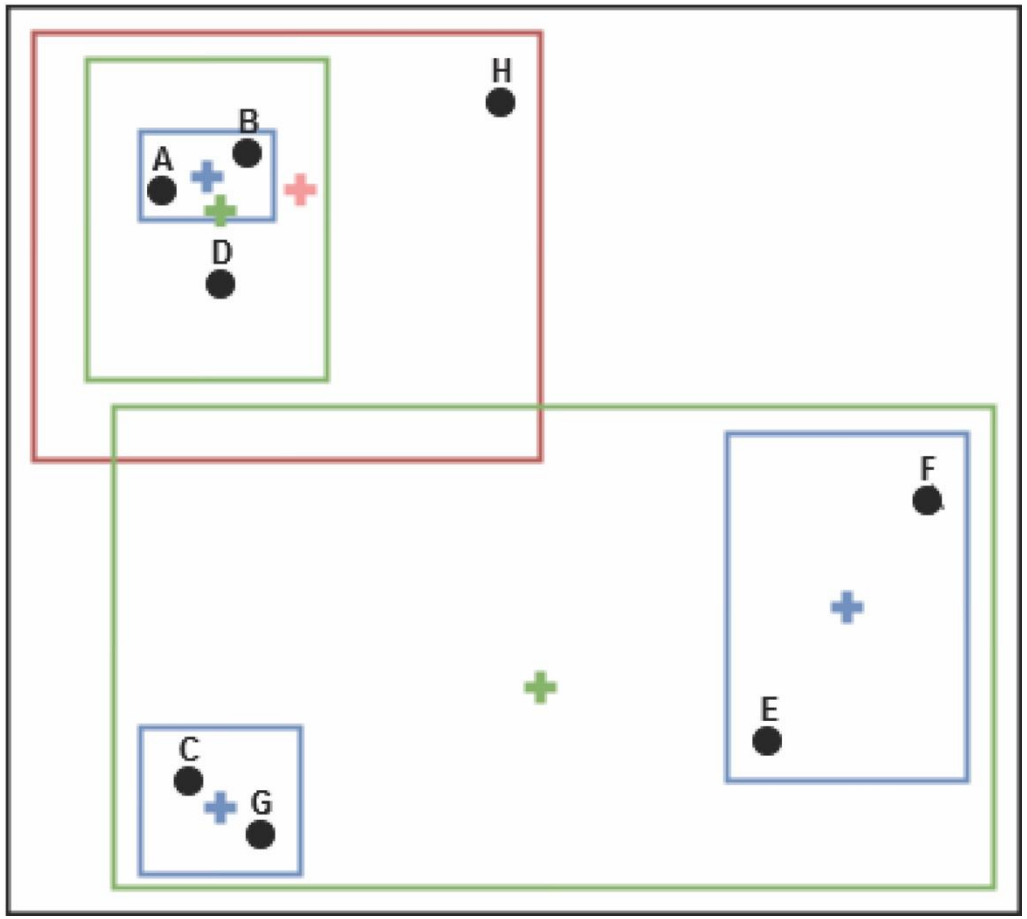
Tree Method – Barnes-Hut





Tree Method – Nearest Neighbour





Optimization

- Cost of pure summation $O(N^2)$
  Highly accurate but expensive

- Tree method reduces cost $O(N \log N)$
  Reduces cost at the expense of accuracy
- Particles grouped into nodes
- Accuracy determined by acceptance criterion

$$r_{ij}^2 > R_{crit}^2$$

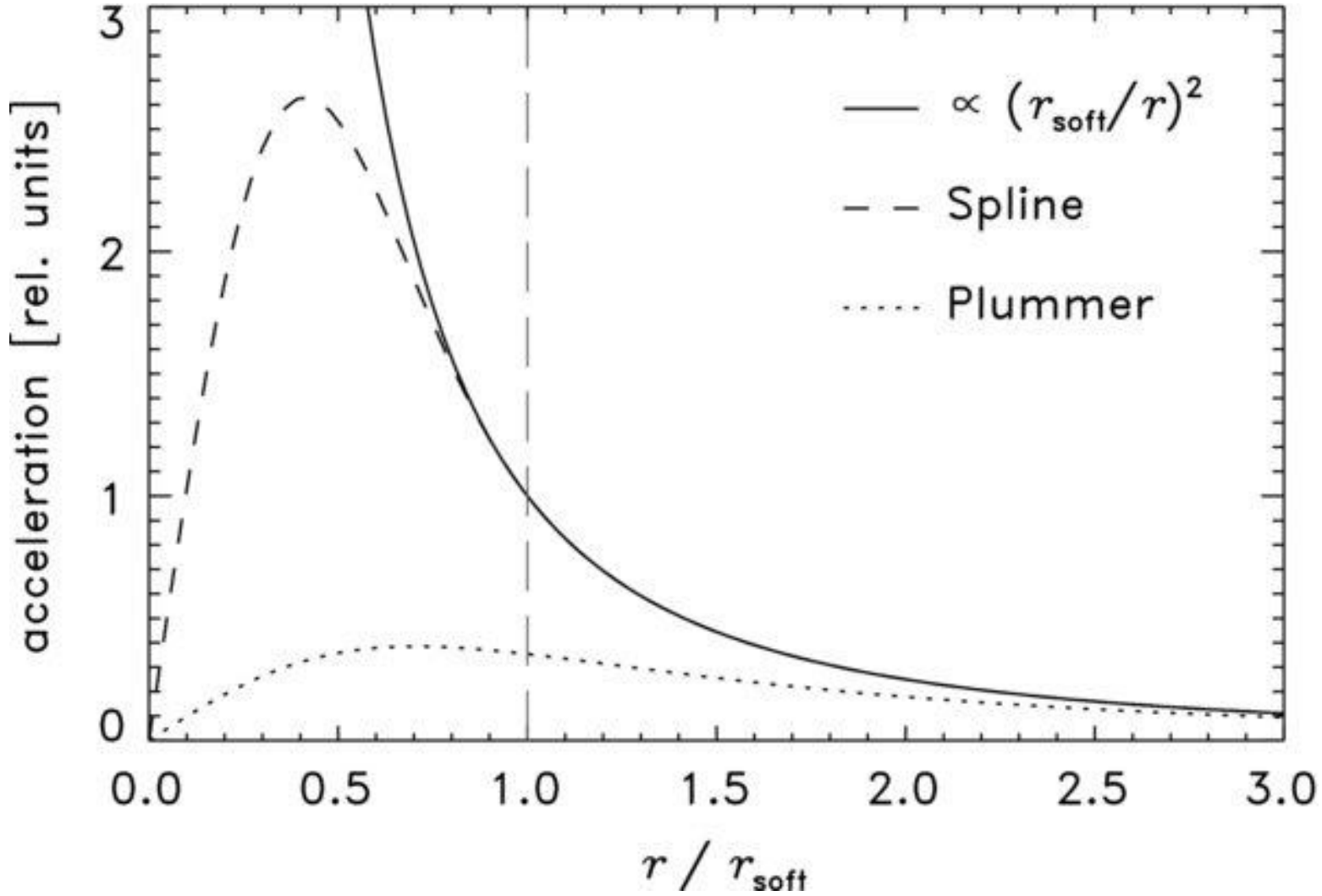
r_{ij} distance between particle and CM of node

Gravitational Softening

- Implemented to avoid divergence

$$F \to \infty \; as \; r_i - r_j \to 0$$

- Soften the potential
  Smooth out the mass



Gravitational Softening

- Implemented to avoid divergence

$$F \to \infty \; as \; r_i - r_j \to 0$$

- Soften the potential
  Smooth out the mass

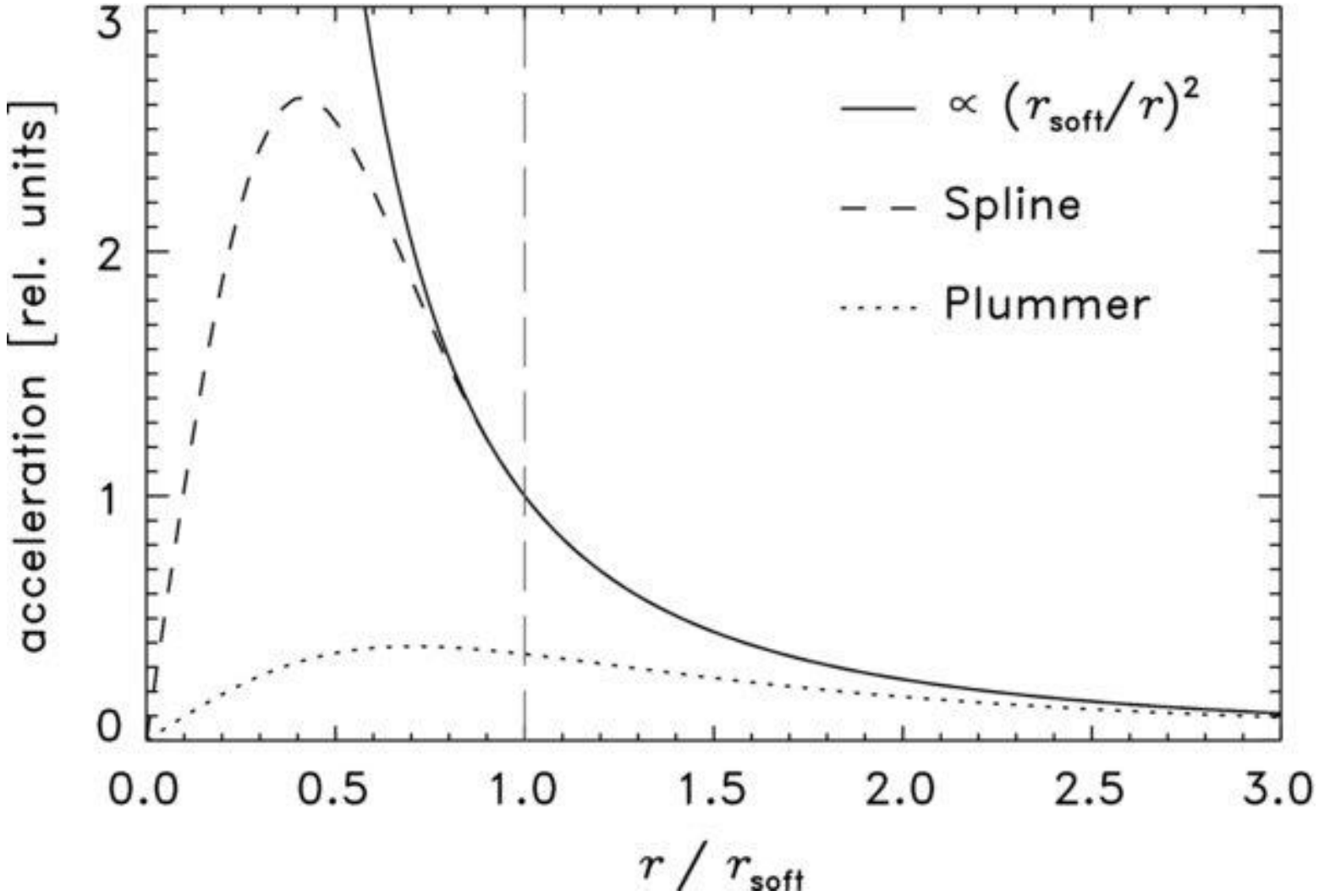
- Plummer model:

$$\delta(r_{ij}) = \frac{1}{\pi} \lim_{\epsilon \to 0} \frac{r_{soft}}{r_{ij}^2 + r_{soft}^2}$$

Type equation here.

$$F_{\mathrm{i}} = -Gm_i \sum_{j \neq i}^{N} m_j \frac{\hat{\mathbf{r}}_{\mathbf{ij}}}{r_{ij}^2 + r_{soft}^2}$$

- r_{soft} = softening parameter



Artificial Viscosity in SPH

- Adds dissipation to the SPH formalism
- Required to handle shocks
- Acts in convergent flows
- Can produce unwanted dissipation in shear flow
- Balsara factor reduces dissipation in shear flow
- AV switch to reduce dissipation away from shocks

